

Δ -KLPT⁺: Improved Norm Bounds for the Quaternion Isogeny Path Problem and an Application to Ring Signatures

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Abstract. Computing smooth-degree isogenies between supersingular elliptic curves with known endomorphism rings is a fundamental task in isogeny-based cryptography. Via the Deuring correspondence, this task is equivalent to finding a connecting ideal of smooth norm between the corresponding quaternion maximal orders. KLPT-type algorithms address this problem by transforming a known connecting ideal into an equivalent ideal of smooth norm. Although connecting ideals of norm approximately p are expected to exist, the best known such algorithms for general input only achieve norm $\tilde{O}(p^{4.5})$. In this paper, we propose Δ -KLPT⁺, a new KLPT-type algorithm that lowers this bound to $\tilde{O}(p^4)$. Our construction builds on Δ -KLPT, which attains a smaller norm bound but requires two input ideals of the same prime norm. To extend Δ -KLPT to arbitrary inputs, we introduce a new subroutine `IdealNormReduce`, which reduces the common norm of an ideal pair. By iterating `IdealNormReduce`, we obtain two ideals with a common norm below p . Then we apply Δ -KLPT to the resulting ideal pair to obtain a desired connecting ideal of smooth norm. As an application, we construct *Delfar*, a fully anonymous isogeny-based ring signature scheme. Compared to Erebor-full, *Delfar* achieves shorter signatures and lower signing and verification costs for large rings.

1 Introduction

Quaternion isogeny path problem. Isogeny-based protocols such as the original SQIsign [17] require computing an isogeny of smooth degree between two supersingular elliptic curves E_1 and E_2 whose endomorphism rings are known. Under the Deuring correspondence [19], these curves correspond to maximal orders $\mathcal{O}_1$ and $\mathcal{O}_2$ of the quaternion algebra $\mathcal{B}_{p,\infty}$, while an isogeny between them corresponds to a connecting ideal from $\mathcal{O}_1$ to $\mathcal{O}_2$ whose norm equals the isogeny degree. In this work, we focus on the case where the smooth degree is a power of a fixed small prime ℓ . Accordingly, the problem considered here can be formulated as follows: given two maximal orders $\mathcal{O}_1$ and $\mathcal{O}_2$ and a small prime ℓ , find a connecting ideal from $\mathcal{O}_1$ to $\mathcal{O}_2$ of norm ℓ^e for some integer e .

This problem is referred to as the *quaternion isogeny path problem* and is highlighted as one of the important problems in isogeny-based cryptography in

[12, Problem 4]. Its optimal variant, called the *optimal quaternion isogeny path problem*, asks for such a connecting ideal with the exponent e of ℓ^e as small as possible. For two generic maximal orders, the minimum such norm is expected to be approximately p , i.e., $\ell^e \approx p$. Unfortunately, no polynomial-time algorithm is currently known for solving the optimal quaternion isogeny path problem. However, when the target norm ℓ^e is sufficiently large, *KLPT-type algorithms* can find a connecting ideal of norm ℓ^e .

KLPT-type algorithms. The original KLPT was proposed by Kohel, Lauter, Petit, and Tignol [23]. This algorithm addresses the special case where one of the maximal orders is a *special extremal order* $\mathcal{O}_0$. More precisely, given a left $\mathcal{O}_0$ -ideal I and a prime-power ℓ^e ,³ it can find an equivalent ideal $J \sim I$ of norm ℓ^e if the target norm ℓ^e is larger than a bound $B = \tilde{O}(p^{3.5})$. A subsequent improvement reduces this bound to $\tilde{O}(p^3)$ [30]. KLPT was subsequently applied to the identification and signature schemes of Galbraith, Petit, and Silva [22], where KLPT is used to transform quaternionic secret information into an equivalent ideal of power-smooth norm and hence into an efficiently computable isogeny. KLPT has also played an important role in establishing reductions between the supersingular isogeny path problem and the endomorphism ring problem [20, 38], both of which are fundamental computational problems underlying the security of isogeny-based cryptography.

To overcome the restriction to special extremal orders, **GeneralizedKLPT** was introduced as a building block of SQIsign [17]. Given a connecting ideal I_1 from a special extremal maximal order $\mathcal{O}_0$ to $\mathcal{O}_1$ of norm N and a connecting ideal I from $\mathcal{O}_1$ to $\mathcal{O}_2$, **GeneralizedKLPT** outputs an equivalent ideal $J \sim I$ of norm ℓ^e . The norm bound of **GeneralizedKLPT** is $\tilde{O}(p^3 N^3)$. For a generic maximal order $\mathcal{O}_1$, one expects a suitable ideal I_1 to have norm approximately $p^{1/2}$, yielding a norm bound of $\tilde{O}(p^{4.5})$ [17, Proposition 9]. Thus, the best known bound for the general quaternion isogeny path problem is $\tilde{O}(p^{4.5})$, which remains far from the expected optimal norm of approximately p .

More recently, Δ -KLPT was proposed as a building block of Δ -SQIsign, which achieves a smaller output norm under additional structural conditions [27]. Given two connecting ideals I_1 and I_2 from $\mathcal{O}_0$ to $\mathcal{O}_1$ and $\mathcal{O}_2$, respectively, of the same prime norm N , Δ -KLPT computes a connecting ideal from $\mathcal{O}_1$ to $\mathcal{O}_2$ of norm ℓ^e . In this setting, Δ -KLPT has a norm bound of $\tilde{O}(pN^3)$, which is smaller than $\tilde{O}(p^3 N^3)$ of **GeneralizedKLPT**. However, Δ -KLPT crucially requires the two input ideals to have the same prime norm. This condition is not available for a generic pair of maximal orders $\mathcal{O}_1$ and $\mathcal{O}_2$, and hence the bound of Δ -KLPT does not directly yield an improvement for the general quaternion isogeny path problem.

Other approaches to the quaternion isogeny path problem. Benčína et al. [8] apply Coppersmith's small-root techniques to quaternion norm equations, obtaining

³ The target norm of KLPT need not be a prime-power, or even smooth; it suffices that it be the product of two sufficiently large integers. Other variants of KLPT likewise do not necessarily require the target norm to be a prime-power. In this work, however, we restrict our attention to the prime-power case.

an algorithm with running time $\tilde{O}(p^{\varepsilon/2})$ for prescribed norms $d \approx p^{1/2+\varepsilon}$ with $0 < \varepsilon < 1/4$. The algorithm finds a connecting ideal of norm d , or determines that none exists. Building on this approach, Aardal et al. [3, Theorem 5] obtain a heuristic polynomial-time algorithm for finding a connecting ideal of prescribed norm $d = p^{1/2+\varepsilon}$, for any fixed $0 < \varepsilon < 1/6$, or determining that none exists. For a generic pair of maximal orders, however, connecting ideals in these norm ranges are expected not to exist with high probability. Thus, these methods do not address the generic case. In addition, Eriksen and Leroux [21] provide a method that can find the smallest prime-power norm with an integer-factorization oracle in heuristic expected time $\tilde{O}(p^{2/3})$, which is exponential in $\log p$.

Application to ring signatures. In recent years, two ring signature schemes based on KLPT-type algorithms have been proposed. Borin et al. constructed the isogeny-based ring signature scheme Erebor [11] by applying the AOS transform [4] to the Σ -protocol of SQIsign, where a variant of **GeneralizedKLPT** is used. Erebor has two variants: Erebor-short achieves shorter signatures but does not provide full anonymity, whereas Erebor-full achieves full anonymity at the cost of longer signatures.

More recently, Nakagawa and Kunihiro constructed a ring signature scheme based on Δ -KLPT [26]. By exploiting the smaller output norm achieved by Δ -KLPT, their construction slightly reduces the computational cost compared with Erebor. However, unlike Erebor-full, it does not achieve full anonymity.

1.1 Contributions

We introduce a new algorithm Δ -KLPT⁺ (Algorithm 2), which solves the quaternion isogeny path problem for arbitrary instances. Our starting point is Δ -KLPT, whose norm bound is smaller than that of **GeneralizedKLPT** under a restricted setting where two input ideals have the same prime norm. To meet this requirement for arbitrary inputs, Δ -KLPT⁺ first aligns the input norms using KLPT, and then repeatedly applies our new subroutine **IdealNormReduce** (Algorithm 3).

The subroutine **IdealNormReduce** is the main technical component of our construction. Given two left $\mathcal{O}_0$ -ideals of the same norm N , it outputs equivalent ideals of the same norm N' satisfying $N' < \delta^{3/2} p^{3/4} N^{1/4}$ for $\delta = 2\sqrt{2}/\pi$. In particular, if $N > \delta^2 p$, then $N' < N$; since $\delta^2 < 1$, this condition is always satisfied when $N > p$. Thus, by iterating this procedure, we can reduce the common norm below p . We then apply Δ -KLPT to the resulting ideal pair to obtain the desired connecting ideal of prime-power norm. Since the norm bound of Δ -KLPT is $\tilde{O}(pN^3)$ and the resulting common norm satisfies $N < p$, our construction achieves a norm bound of $\tilde{O}(p^4)$, improving upon the previous bound of $\tilde{O}(p^{4.5})$ achieved by **GeneralizedKLPT**.

Application to ring signatures. For a concrete cryptographic application, we develop Δ -SigningKLPT⁺, a signing variant of Δ -KLPT⁺ that additionally ensures

the cyclicity condition required for signing. Using this variant, we construct *Delfar*, a new isogeny-based ring signature scheme with full anonymity. As in Erebor-full [11], we construct a Σ -protocol and then apply the AOS transform [4]. The main difference lies in response generation: Delfar uses Δ -SigningKLPT⁺, whereas Erebor-full uses a modified version of GeneralizedKLPT. The smaller norm bound of Δ -SigningKLPT⁺ allows Delfar to achieve shorter signatures and lower signing and verification costs for large rings than Erebor-full.

1.2 Organization

The remainder of this paper is organized as follows. Section 2 reviews some background knowledge used in this paper. Section 3 presents IdealNormReduce, Δ -KLPT⁺, and its signing variant Δ -SigningKLPT⁺, together with their analyses. Section 4 describes the ring signature scheme Delfar and analyzes its security and efficiency. Finally, Section 5 gives directions for future work.

2 Preliminaries

In this section, we summarize some background knowledge used in this paper.

2.1 Notation

Throughout this paper, we use the following notation. We let p be a prime of cryptographic size (at least approximately 2^{256}), and let λ be a security parameter. For a finite set S , we write $x \xleftarrow{\$} S$ if x is sampled uniformly at random from S . For a positive integer n , we define $[n] := \{1, \dots, n\}$. For an odd prime q and an integer a , $\left(\frac{a}{q}\right)$ denotes the Legendre symbol. Throughout the paper, $\log$ denotes the natural logarithm. A function $f(\lambda)$ is *negligible* if $f(\lambda) = o(\lambda^{-\alpha})$ for every constant $\alpha > 0$. We say that an event occurs with *overwhelming probability* if its failure probability is negligible.

2.2 Elliptic Curves and Isogenies

We assume familiarity with the basic theory of elliptic curves and isogenies and refer the reader to [34] for more details. Throughout this paper, we consider supersingular elliptic curves over finite fields of characteristic p . Since every supersingular elliptic curve admits a model over $\mathbb{F}_{p^2}$, we restrict our attention to elliptic curves defined over $\mathbb{F}_{p^2}$.

An *isogeny* is a non-constant rational map between elliptic curves that is also a group homomorphism. The *degree* of an isogeny φ is its degree as a rational map and is denoted by $\deg \varphi$. When $\deg(\varphi) = \ell$ for an integer ℓ , we say that φ is an ℓ -*isogeny*. An isogeny φ is *separable* if $\#\ker \varphi = \deg \varphi$. A separable isogeny is uniquely determined by its kernel up to post-composition by an isomorphism. A separable isogeny is called *cyclic* if its kernel is cyclic. For an isogeny φ :

$E \rightarrow E'$, there exists an isogeny $\widehat{\varphi} : E' \rightarrow E$ called the *dual isogeny*, satisfying $\widehat{\varphi} \circ \varphi = [\deg \varphi]$ and $\varphi \circ \widehat{\varphi} = [\deg \varphi]$.

For a positive integer ℓ , we denote by $E[\ell]$ the ℓ -torsion subgroup of E . Let E be an elliptic curve defined over $\mathbb{F}_{p^2}$ and let ℓ be an integer such that $E[\ell] \subseteq E(\mathbb{F}_{p^2})$. Then we can compute a deterministic basis of $E[\ell]$ using deterministic algorithms such as [13, Algorithm 3] and [2, Algorithm 2.2]. If the p^2 -power Frobenius map of E is $[-p]$, then $E[m] \subseteq E(\mathbb{F}_{p^2})$ for every m dividing $p + 1$. Moreover, this Frobenius condition is preserved under isogenies defined over $\mathbb{F}_{p^2}$. Consequently, for the parameters $p = c \cdot 2^f - 1$ used in Section 4, the curve E_0 introduced below has the p^2 -power Frobenius $[-p]$, and the 2^f -torsion of E_0 and of every curve reached from E_0 by an $\mathbb{F}_{p^2}$ -rational isogeny is rational over $\mathbb{F}_{p^2}$.

Given $K \in E[\ell^e]$ for a small prime ℓ and a positive integer e , we can compute an ℓ^e -isogeny $\varphi : E \rightarrow E'$ with kernel $\langle K \rangle$ in polynomial time by using *Vélu's formulas* [36].

2.3 Quaternion Algebras and the Deuring Correspondence

We briefly recall the basic notions of quaternion algebras and the Deuring correspondence needed in this paper; for more details, we refer the reader to [37].

Quaternion algebras. A *quaternion algebra* over $\mathbb{Q}$ is a four-dimensional $\mathbb{Q}$ -algebra $H(a, b) = \mathbb{Q} \oplus \mathbb{Q}\mathbf{i} \oplus \mathbb{Q}\mathbf{j} \oplus \mathbb{Q}\mathbf{k}$ defined by the relations $\mathbf{i}^2 = a$, $\mathbf{j}^2 = b$, and $\mathbf{ij} = -\mathbf{ji} = \mathbf{k}$, where $a, b \in \mathbb{Q}^*$. We say $H(a, b)$ is *ramified* at a place v of $\mathbb{Q}$ if $H(a, b) \otimes_{\mathbb{Q}} \mathbb{Q}_v$ is not isomorphic to the matrix algebra $M_2(\mathbb{Q}_v)$. There exists a quaternion algebra ramified exactly at p and ∞ . Such an algebra is unique up to isomorphism. We denote it by $\mathcal{B}_{p, \infty}$.

Let $\alpha = x + y\mathbf{i} + z\mathbf{j} + t\mathbf{k} \in H(a, b)$ with $x, y, z, t \in \mathbb{Q}$. The *conjugate* of α is $x - y\mathbf{i} - z\mathbf{j} - t\mathbf{k}$ and is denoted by $\bar{\alpha}$. The (*reduced*) *norm* of α is $\alpha\bar{\alpha}$ and is denoted by $\text{nrd}(\alpha)$.

A full-rank $\mathbb{Z}$ -lattice $\mathcal{O}$ in $H(a, b)$ is called an *order* if it is a subring of $H(a, b)$. An order $\mathcal{O}$ is said to be *maximal* if there is no larger order that contains $\mathcal{O}$. For a full-rank $\mathbb{Z}$ -lattice I of $H(a, b)$, we define $\mathcal{O}_L(I) := \{\alpha \in H(a, b) \mid \alpha I \subseteq I\}$. This is an order of $H(a, b)$, called the *left order* of I . The *right order* $\mathcal{O}_R(I)$ of I is defined similarly. For a maximal order $\mathcal{O}$, a *left $\mathcal{O}$ -ideal* I is a full-rank $\mathbb{Z}$ -lattice such that $\mathcal{O}_L(I) = \mathcal{O}$. A *right $\mathcal{O}$ -ideal* is similarly defined. When I is a left $\mathcal{O}$ -ideal and right $\mathcal{O}'$ -ideal, we say that I is a *connecting ideal* from $\mathcal{O}$ to $\mathcal{O}'$.

For an ideal I , we denote its conjugate by $\bar{I} = \{\bar{\alpha} \mid \alpha \in I\}$. We denote by $\text{nrd}(I)$ the (*reduced*) *norm* of the ideal I , defined as (the unique positive generator of) the $\mathbb{Z}$ -module generated by the norms of the elements of I . A left $\mathcal{O}$ -ideal I of integer norm can be written as $I = \mathcal{O}\alpha + \mathcal{O}\text{nrd}(I)$ for some $\alpha \in I$. Ideals I and J are *equivalent*, written $I \sim J$, if there exists $\beta \in H(a, b)^*$ such that $J = I\beta$. If I is a connecting ideal from $\mathcal{O}$ to $\mathcal{O}'$, then J is a connecting ideal from $\mathcal{O}$ to $\beta^{-1}\mathcal{O}'\beta$, which is isomorphic to $\mathcal{O}'$ as a ring. A left $\mathcal{O}$ -ideal I is called *primitive* if $I \not\subseteq m\mathcal{O}$ for every integer $m > 1$.

If I is a connecting ideal from $\mathcal{O}$ to $\mathcal{O}'$ and J is a connecting ideal from $\mathcal{O}'$ to a maximal order $\mathcal{O}''$, then the product IJ is a connecting ideal from $\mathcal{O}$ to $\mathcal{O}''$ of norm $\text{nr}(IJ) = \text{nr}(I)\text{nr}(J)$. We write $I^{-1} := \frac{\bar{I}}{\text{nr}(I)}$. Then I^{-1} is a connecting ideal from $\mathcal{O}'$ to $\mathcal{O}$ and satisfies $II^{-1} = \mathcal{O}$ and $I^{-1}I = \mathcal{O}'$.

We also recall the push-forward and pull-back operations on ideals [13, Section 2.4.5.1]. Let I be a connecting ideal from $\mathcal{O}$ to $\mathcal{O}'$. For a left $\mathcal{O}$ -ideal J with $\gcd(\text{nr}(I), \text{nr}(J)) = 1$, its *push-forward* by I is the left $\mathcal{O}'$ -ideal $[I]_*J := I^{-1}(I \cap J)$. Conversely, for a left $\mathcal{O}'$ -ideal K with $\gcd(\text{nr}(I), \text{nr}(K)) = 1$, its *pull-back* by I is the left $\mathcal{O}$ -ideal $[I]^*K := IK + \text{nr}(K)\mathcal{O}$.

The following lemma is a direct consequence of Minkowski's theorem and follows by the same argument as in [16, Lemma 12].

Lemma 1. *Let L be a full-rank $\mathbb{Z}$ -lattice of index N in a maximal order $\mathcal{O}$ of $\mathcal{B}_{p,\infty}$. Then, there exists a nonzero element $\alpha \in L$ such that $\text{nr}(\alpha) \leq \frac{2\sqrt{2}}{\pi}\sqrt{pN}$.*

To ease the notation, we denote by δ the factor $\frac{2\sqrt{2}}{\pi}$ in the above lemma. Note that $\delta < 1$.

Deuring correspondence. Deuring [19] showed that the endomorphism ring of a supersingular elliptic curve over $\mathbb{F}_{p^2}$ is isomorphic to a maximal order of $\mathcal{B}_{p,\infty}$. The resulting *Deuring correspondence* associates a supersingular elliptic curve E over $\mathbb{F}_{p^2}$ with a maximal order isomorphic to $\text{End}(E)$.

Suppose that $p \equiv 3 \pmod{4}$, as in our scheme. Then we can take $\mathcal{B}_{p,\infty} = H(-1, -p)$ and the elliptic curve $E_0/\mathbb{F}_{p^2} : y^2 = x^3 + x$ is supersingular. We define endomorphisms $\iota : (x, y) \mapsto (-x, \sqrt{-1}y)$ and $\pi_p : (x, y) \mapsto (x^p, y^p)$ of E_0 , where $\sqrt{-1}$ is a fixed square root of -1 in $\mathbb{F}_{p^2}$ and π_p is the p -th power Frobenius map satisfying $\pi_p^2 = -p$. Then $\text{End}(E_0)$ is isomorphic to $\mathcal{O}_0 := \mathbb{Z}\langle 1, \mathbf{i}, \frac{1+\mathbf{j}}{2}, \frac{1+\mathbf{k}}{2} \rangle$, which is a *special extremal order* defined in [23]. This isomorphism is given by $\iota \mapsto \mathbf{i}$ and $\pi_p \mapsto \mathbf{j}$. Throughout this paper, $\mathcal{O}_0$ denotes this maximal order.

The Deuring correspondence also establishes a correspondence between isogenies and ideals. Let $\varphi : E_1 \rightarrow E_2$ be a d -isogeny between supersingular elliptic curves over $\mathbb{F}_{p^2}$, and let $\mathcal{O}_1$ and $\mathcal{O}_2$ be the maximal orders of $\mathcal{B}_{p,\infty}$ corresponding to E_1 and E_2 , respectively. Let $\iota_1 : \mathcal{O}_1 \xrightarrow{\sim} \text{End}(E_1)$ be an isomorphism. Then the isogeny φ can be associated with a connecting ideal I_φ from $\mathcal{O}_1$ to $\mathcal{O}_2$ of norm d defined by

$$I_\varphi := \{\alpha \in \mathcal{O}_1 \mid \iota_1(\alpha)(P) = 0, \forall P \in \ker \varphi\}.$$

Conversely, for a connecting ideal I from $\mathcal{O}_1$ to $\mathcal{O}_2$ of norm d , we can define the corresponding isogeny $\varphi_I : E_1 \rightarrow E_2$ by setting its kernel to

$$E_1[I] := \{P \in E_1 \mid \iota_1(\alpha)(P) = 0, \forall \alpha \in I\}.$$

Two isogenies $\varphi, \psi : E_1 \rightarrow E_2$ with the same domain and codomain correspond to equivalent ideals $I_\varphi \sim I_\psi$. For a connecting ideal I from $\mathcal{O}_1$ to $\mathcal{O}_2$ and $\alpha \in I \setminus \{0\}$, we define $\chi_I(\alpha) := I \frac{\bar{\alpha}}{\text{nr}(I)}$, which is an ideal equivalent to I of norm $\frac{\text{nr}(\alpha)}{\text{nr}(I)}$.

Methods for computing an isogeny corresponding to a given ideal have been studied extensively [20, 17, 18, 6, 28, 10]. In this paper, we use the algorithm

proposed in [10], which converts a left $\mathcal{O}_0$ -ideal I of odd norm N to the corresponding isogeny φ_I of degree N . More precisely, this algorithm is designed for primes of the form $p = c \cdot 2^f - 1$, and it returns the action of φ_I on $E_0[2^f]$.⁴ We denote this algorithm by $\text{IdealTolsogeny}(I)$.

2.4 KLPT-Type Algorithms and Related Subroutines

We recall some KLPT-type algorithms and related subroutines. From now on, we let e be a positive integer and ℓ be a small prime distinct from p .

$\text{KLPT}_{\ell^e}(I)$: This algorithm takes as input a left $\mathcal{O}_0$ -ideal I and a prime-power ℓ^e , and outputs an equivalent ideal $J \sim I$ of norm ℓ^e . KLPT was later applied to identification and signature schemes in [22], as well as to reductions between the supersingular isogeny path problem and the endomorphism ring problem [20, 38]. For the construction of KLPT, Kohel et al. also introduced several building blocks such as `RepresentInteger`, `IdealModConstraint`, and `StrongApproximation`. We now recall `IdealModConstraint` and `StrongApproximation`, which are explicitly used in this work.

- `IdealModConstraint`(I_1, β_2): It takes as input a left $\mathcal{O}_0$ -ideal I_1 of prime norm $N \neq p$ and an element $\beta_2 \in \mathcal{O}_0$ such that $\gcd(N^2, \text{nrd}(\beta_2)) = N$, and outputs $C, D \in \mathbb{Z}$ such that $\beta_2(C\mathbf{j} + D\mathbf{k}) \in I_1$. In this paper, for an ideal $I_2 = \mathcal{O}_0\langle\beta_2, N\rangle$ of norm N , we write `IdealModConstraint`(I_1, I_2) for the invocation `IdealModConstraint`(I_1, β_2). In this case, its output satisfies $I_2(C\mathbf{j} + D\mathbf{k}) \subseteq I_1$. If I_1 and I_2 are sampled uniformly at random among all left $\mathcal{O}_0$ -ideals of norm N , this algorithm succeeds with probability $1 - 4(N-1)/(N+1)^2$ when $N \equiv 1 \pmod{4}$, and always succeeds when $N \equiv 3 \pmod{4}$; see [24, Lemma 3.2.6].
- `StrongApproximation` $_{\ell^e}(N, C, D)$: It takes as input a prime power ℓ^e , a prime $N \neq p, \ell$, and $(C : D) \in \mathbb{P}^1(\mathbb{Z}/N\mathbb{Z})$ satisfying $(\frac{p\ell^e(C^2+D^2)}{N}) = 1$, and outputs an element $\alpha \in \mathcal{O}_0$ of norm ℓ^e of the form $\alpha = u(C\mathbf{j} + D\mathbf{k}) + N\mu$, where $u \in \mathbb{Z}$ and $\mu \in \mathcal{O}_0$. If $(C : D)$ is the output of `IdealModConstraint`(I_1, I_2), α satisfies $I_2\alpha \subseteq I_1$.

The search of μ performed by `StrongApproximation` is restricted to the sub-order $\mathbb{Z}[\mathbf{i}, \mathbf{j}] \subsetneq \mathcal{O}_0$. De Feo, Leroux, Longa, and Wesolowski [18] proposed the variant `FullStrongApproximation`, which instead searches over the full maximal order $\mathcal{O}_0$. The larger search space increases the probability of finding a solution.

Originally, `(Full)StrongApproximation` required a norm ℓ^e above a bound of $\tilde{O}(pN^4)$ to succeed with overwhelming probability. Petit and Smith [30] improved this norm bound to $\tilde{O}(pN^3)$. Consequently, the norm bound achieved

⁴ Provided that $N < 2^f$, this restriction $\varphi_I|_{E_0[2^f]}$, together with the degree N , suffices to evaluate φ_I at arbitrary points. This follows from Robert's higher-dimensional representation of isogenies [32, Theorem 5.19].

by KLPT was also improved from $\tilde{O}(p^{3.5})$ to $\tilde{O}(p^3)$. Throughout this paper, we use `FullStrongApproximation` and KLPT incorporating this improvement. To analyze the norm bounds of our algorithm Δ -KLPT⁺ later, we use the following heuristics for `FullStrongApproximation` and KLPT.

Heuristic 2 *Let N be a prime such that $N \neq p, \ell$ and $\omega > 0$. Suppose that ℓ^e, N, C, D satisfy $\left(\frac{p\ell^e(C^2+D^2)}{N}\right) = 1$. If*

$$\ell^e > \frac{3\log(2)}{2\pi}\omega \cdot pN^3 \log(pN^3)$$

then `FullStrongApproximation` _{ℓ^e} (N, C, D) succeeds with probability at least $1 - 2^{-\omega}$ and runs in expected time polynomial in $\log p$, $\log N$, and ω .

Heuristic 3 *For a positive integer ω , there exists a bound*

$$B_{\text{KLPT}}(p, \omega) = 3\log_2 p + O(\log \log p) + O(\log \omega)$$

such that, if $e_0 > B_{\text{KLPT}}(p, \omega)$, then `KLPT` _{2^{e_0}} succeeds with probability at least $1 - 2^{-\omega}$ and runs in expected time polynomial in $\log p$ and ω .

These heuristics refine the norm bounds for `FullStrongApproximation` and KLPT discussed above. A concrete derivation of the bound in Heuristic 2 is given in Appendix B. Heuristic 3 is a natural heuristic consequence of [24, Proposition 3.2.7].

GeneralizedKLPT _{ℓ^e} (I_1, I): This algorithm was proposed as a building block of the original SQIsign [17]. Let I_1 be a connecting ideal from $\mathcal{O}_0$ to $\mathcal{O}_1$ of norm N , and let I be a connecting ideal from $\mathcal{O}_1$ to $\mathcal{O}_2$ whose norm is coprime to N . Given I_1, I , and a prime power ℓ^e , the algorithm outputs an equivalent ideal $J \sim I$ whose norm divides ℓ^e . The norm bound for **GeneralizedKLPT** is $\tilde{O}(p^3 N^3)$. For general inputs, one can take I_1 with norm approximately $p^{1/2}$, and hence this bound becomes $\tilde{O}(p^{4.5})$ [17, Proposition 9]. The variant that modifies the algorithm so that the output ideal has norm exactly ℓ^e is called **SigningKLPT** and was also proposed in [17].

Δ -KLPT _{ℓ^e} (I_1, I_2): For two left $\mathcal{O}_0$ -ideals I_1 and I_2 of the same prime norm N , we define

$$\text{QR}_{p, \ell^e}(I_1, I_2) := \left(\frac{p\ell^e(C^2 + D^2)}{N} \right), \text{ for } (C : D) \leftarrow \text{IdealModConstraint}(I_1, I_2),$$

which is uniquely determined by I_1 and I_2 . Δ -KLPT takes as input two left $\mathcal{O}_0$ -ideals I_1 and I_2 of the same norm N such that N is a prime congruent to 3 modulo 4 and $\text{QR}_{p, \ell^e}(I_1, I_2) = 1$. Then it outputs an equivalent ideal $I \sim \overline{I_1}I_2$ of norm ℓ^e . The algorithm proceeds as follows:

1. Let $(C : D) \leftarrow \text{IdealModConstraint}(I_1, I_2)$.

Algorithm 1: AOS Sequential Ring Signature [4]

```

KeyGen( $1^\lambda$ );
1 (pk, sk)  $\leftarrow$  Gen( $1^\lambda$ );
2 return (pk, sk);

Sign( $sk_l, m, R$ );
1 ( $com_l, st_l$ )  $\leftarrow$  P1(pkl, skl);
2 chll+1  $\leftarrow$  H( $com_l, R, m, pk_{l+1}$ );
3 for  $i = l + 1, \dots, n, 1, \dots, l - 1$  do
4   ( $com_i, rsp_i$ )  $\leftarrow$  S(pki, chli);
5   chli+1  $\leftarrow$  H( $com_i, R, m, pk_{i+1}$ );
6 rspl  $\leftarrow$  P2(pkl, skl, stl, chll);
7  $\sigma \leftarrow$  (chl1, com1, ..., comn,;
8   rsp1, ..., rspn);
9 return  $\sigma$ ;

Verify( $m, R, \sigma$ );
1 for  $i = 1, \dots, n$  do
2   if V(pki, comi, chli, rspi) = 0 then
3     return reject;
4   chli+1  $\leftarrow$  H( $com_i, R, m, pk_{i+1}$ );
5 if chln+1 = chl1 then
6   return accept;
7 else
8   return reject;

```

2. Compute $\alpha \leftarrow \text{FullStrongApproximation}_{\ell^e}(N, C, D)$. $// I_2 \alpha \subseteq I_1$.
3. Output $I = I_1^{-1} I_2 \alpha$.

Since Δ -KLPT invokes FullStrongApproximation, the target norm ℓ^e must exceed a bound of $\tilde{O}(pN^3)$. Note that this algorithm is based on the following diagram. For readability, the diagram is drawn up to isomorphism.⁵

$$\begin{array}{ccc}
 \mathcal{O}_0 & \xrightarrow{\mathcal{O}_0 \alpha} & \mathcal{O}_0 \\
 I_1 \downarrow & & \downarrow I_2 \\
 \mathcal{O}_1 & \xrightarrow{I} & \mathcal{O}_2
 \end{array}$$

2.5 Ring Signatures from Σ -Protocols

We give the definitions of Σ -protocols and ring signatures in Appendix A. Ring signatures can be constructed from Σ -protocols using the AOS transform [4]. Let $\Pi = (\text{Gen}, P_1, P_2, V)$ be a special sound and special HVZK Σ -protocol for a hard relation $\mathcal{R}$, with challenge space $\mathcal{C}$ and a PPT simulator $\mathcal{S}$. Let $H : \{0, 1\}^* \rightarrow \mathcal{C}$ be a hash function, modeled as a random oracle [7]. The resulting ring signature scheme is given in Algorithm 1.

The following proposition establishes the security of the ring signature obtained via the AOS transform.

Proposition 4 ([11, Proposition 1, Proposition 2]). *In the random oracle model, if the underlying Σ -protocol for a hard relation is correct, special sound,*

⁵ For example, the right order of the principal ideal $\mathcal{O}_0 \alpha$ is strictly speaking $\alpha^{-1} \mathcal{O}_0 \alpha$, which is identified in the diagram with the isomorphic order $\mathcal{O}_0$.

and special (strong) HVZK, then the ring signature scheme obtained via the AOS transform is correct, unforgeable with respect to insider corruption, and (fully) anonymous.

2.6 SQIsign and Erebor

Here, we recall the signature scheme SQIsign and the ring signature scheme Erebor.

SQIsign. SQIsign is an isogeny-based signature scheme proposed by De Feo, Kohel, Leroux, Petit, and Wesolowski in [17]. SQIsign is obtained by applying the Fiat–Shamir transform to the Σ -protocol based on the following diagram.

$$\begin{array}{ccc} E_0 & \xrightarrow{\varphi_{\text{sk}}} & E_{\text{pk}} \\ \varphi_{\text{com}} \downarrow & & \downarrow \varphi_{\text{chl}} \\ E_{\text{com}} & \xrightarrow{\varphi_{\text{rsp}}} & E_{\text{chl}} \end{array}$$

Note that there are multiple variants of SQIsign, among which isogeny φ_{chl} may originate from E_{pk} or E_{com} . To be consistent with the formulations of Erebor and Delfar, we describe the variant in which φ_{chl} originates from E_{pk} .

The procedure of the Σ -protocol is as follows.

- **Key generation.** Compute a random isogeny $\varphi_{\text{sk}} : E_0 \rightarrow E_{\text{pk}}$ and its corresponding ideal I_{sk} . Set $\text{pk} = E_{\text{pk}}$ and $\text{sk} = (\varphi_{\text{sk}}, I_{\text{sk}})$.
- **Commitment.** Compute a random isogeny $\varphi_{\text{com}} : E_0 \rightarrow E_{\text{com}}$ and its corresponding ideal I_{com} . Send E_{com} as a commitment.
- **Challenge.** Sample a random isogeny $\varphi_{\text{chl}} : E_{\text{pk}} \rightarrow E_{\text{chl}}$ and send it as a challenge.
- **Response.** Let I_{chl} be the ideal corresponding to φ_{chl} and let $I = \bar{I}_{\text{com}} \cdot I_{\text{sk}} \cdot I_{\text{chl}}$. Compute $I_{\text{rsp}} \leftarrow \text{SigningKLPT}_{2^{e_{\text{rsp}}}}(I_{\text{com}}, I)$ and return its corresponding isogeny $\varphi_{\text{rsp}} : E_{\text{com}} \rightarrow E_{\text{chl}}$ as a response.
- **Verification.** Check that φ_{rsp} is an isogeny from E_{com} to E_{chl} .

Erebor. Erebor is an isogeny-based ring signature proposed by Borin, Lai, and Leroux [11]. Roughly speaking, Erebor is obtained by applying the AOS transform to the above Σ -protocol. To apply the AOS transform, we need a simulator $\mathcal{S}$ for a special HVZK, which can be constructed as follows.

- $\mathcal{S}(\text{pk} := E_{\text{pk}}, \text{chl} := \varphi_{\text{chl}}) \rightarrow (\text{com}, \text{rsp})$:
 1. Compute $\varphi_{\text{chl}} : E_{\text{pk}} \rightarrow E_{\text{chl}}$.
 2. Sample a random $2^{e_{\text{rsp}}}$ -isogeny $\varphi : E_{\text{chl}} \rightarrow E_{\text{com}}$.
 3. Return $(\text{com} = E_{\text{com}}, \text{rsp} = \widehat{\varphi})$.

Erebor has two variants, Erebor-full and Erebor-short. In Erebor-full, to achieve full anonymity, the ideal I_{com} corresponding to φ_{com} is randomized by sampling a random ideal I' equivalent to I_{com} . This randomization ensures that the response ideal I_{rsp} output by $\text{SigningKLPT}_{2^{e_{\text{rsp}}}}(I', I)$ is sufficiently randomized, which leads to full anonymity. In this randomization, I' is sampled from ideals of norm approximately $p^{1/2} \cdot 2^{\lambda/2}$. Consequently, the degree of the response isogeny φ_{rsp} is bounded by $\tilde{O}\left(p^3 \left(p^{1/2} \cdot 2^{\lambda/2}\right)^3\right) = \tilde{O}\left(p^{9/2} \cdot 2^{3\lambda/2}\right)$.

On the other hand, Erebor-short reduces the signature size but does not achieve full anonymity. In Erebor-short, the degree of φ_{com} is chosen to be approximately $p^{1/4}$. Therefore, the degree of φ_{rsp} is reduced to $\tilde{O}(p^{3.75})$.

3 New Algorithm $\Delta\text{-KLPT}^+$

In this section, we present the algorithm $\Delta\text{-KLPT}^+$ (Algorithm 2), which is a novel variant of GeneralizedKLPT [17]. Given arbitrary left $\mathcal{O}_0$ -ideals I_1 and I_2 and a prime power ℓ^e , our goal is to find an ideal $J \sim \overline{I_1} I_2$ of norm ℓ^e . Let $\mathcal{O}_1$ and $\mathcal{O}_2$ be the right orders of I_1 and I_2 , respectively. Then J is a connecting ideal from $\mathcal{O}_1$ to $\mathcal{O}_2' \cong \mathcal{O}_2$.

The starting point is $\Delta\text{-KLPT}$ whose inputs must satisfy restrictive conditions. More precisely, $\Delta\text{-KLPT}_{\ell^e}$ requires that the input ideals I_1 and I_2 satisfy the following conditions:

$$\begin{aligned} \text{Condition 1: } & \text{nrd}(I_1) = \text{nrd}(I_2) =: N, \\ \text{Condition 2: } & N \text{ is prime congruent to 3 modulo 4,} \\ \text{Condition 3: } & \text{QR}_{p, \ell^e}(I_1, I_2) = 1, \end{aligned} \tag{1}$$

where $\text{QR}_{p, \ell^e}(I_1, I_2)$ denotes the quadratic-residue character defined in Subsection 2.4. To extend $\Delta\text{-KLPT}$ to arbitrary inputs, we need to transform two arbitrary ideals I_1 and I_2 into equivalent ideals satisfying these conditions.

A natural approach to achieve Condition 1 is to apply KLPT separately to I_1 and I_2 with the same target norm N , thereby obtaining equivalent ideals of the same norm. However, this approach raises two issues. Firstly, the norm bound of KLPT places N on the scale of $\tilde{O}(p^3)$. Thus, applying $\Delta\text{-KLPT}$ would give a norm bound of $\tilde{O}(pN^3) = \tilde{O}(p^{10})$, already far larger than the bound achieved by GeneralizedKLPT . Secondly, KLPT does not allow a prime target norm,⁶ so its outputs do not satisfy Condition 2. Moreover, the resulting pair of ideals is not guaranteed to satisfy Condition 3.

To address the first issue, we introduce a novel subroutine **IdealNormReduce** (Algorithm 3) that reduces the common norm of an ideal pair while preserving the equivalence class of each ideal. Using this subroutine, we obtain an ideal pair with common norm below p . To address the second issue, we randomize the ideal pair before applying **IdealNormReduce** and repeat this randomization-and-reduction process until the resulting pair satisfies Conditions 2 and 3.

⁶ This restriction can be avoided if the target norm is chosen sufficiently large.

Applying Δ -KLPT to the resulting ideal pair then yields a norm bound of $\tilde{O}(p^4)$, improving upon the best bound of $\tilde{O}(p^{4.5})$ achieved by GeneralizedKLPT.

3.1 Overview

Here, we give an overview of Δ -KLPT⁺. Given left $\mathcal{O}_0$ -ideals I_1 and I_2 and a prime power ℓ^e , Δ -KLPT⁺ finds an ideal $J \sim \overline{I_1}I_2$ of norm ℓ^e . We also fix a parameter $\omega > 0$ that controls the success probability of Δ -KLPT⁺. At a high level, Δ -KLPT⁺ consists of the following four phases:

1. **Norm Alignment.** We transform two input ideals I_1 and I_2 into equivalent ideals of the same norm, thus satisfying Condition 1. We first fix an integer e_0 larger than a bound $B_{\text{KLPT}}(p, \omega)$ defined in Heuristic 3. We then apply $\text{KLPT}_{2^{e_0}}$ independently to I_1 and I_2 to obtain equivalent ideals L_1 and L_2 satisfying $\text{nrd}(L_1) = \text{nrd}(L_2) = 2^{e_0}$.
2. **Randomization.** We randomize the pair of ideals while preserving the equality of their norms. We sample $r \xleftarrow{\$} \{0, \dots, p\}$ and set $\gamma = 1 + r\mathbf{i}$ if $r < p$, and $\gamma = \mathbf{i}$ otherwise. We then set $J_1 = L_1\gamma$ and $J_2 = L_2\bar{\gamma}$. Then we have $\text{nrd}(J_1) = \text{nrd}(J_2) = 2^{e_0} \text{nrd}(\gamma)$. We discuss the effect of this randomization in detail in Subsection 3.3.
3. **Norm Reduction.** We reduce the common norm of the ideal pair. This is the most challenging phase and the key to the improved norm bound. To realize this phase, we construct a new subroutine `IdealNormReduce`, which takes as input two left $\mathcal{O}_0$ -ideals J_1 and J_2 of the same norm N and outputs equivalent ideals J'_1 and J'_2 of the same norm N' , where N' is smaller than $\delta^{3/2}p^{3/4}N^{1/4}$. We repeatedly apply `IdealNormReduce` until the common norm becomes smaller than p . After reducing the norm, we check whether the resulting ideal pair satisfies the remaining requirements of Δ -KLPT $_{\ell^e}$, namely, Conditions 2 and 3. If not, we return to the randomization phase. We give the details of `IdealNormReduce` in Subsection 3.2.
4. **Apply Δ -KLPT.** Finally, we apply Δ -KLPT $_{\ell^e}$ to the resulting pair (J_1, J_2) . Note that the call to `IdealModConstraint` within Δ -KLPT can be omitted here, because `IdealModConstraint` has already been invoked to check Condition 3. The direct output $\tilde{J} = J_1^{-1}J_2\alpha$ of Δ -KLPT satisfies $\tilde{J} \sim \overline{J_1}J_2$, but need not satisfy $\tilde{J} \sim \overline{I_1}I_2$. In fact, for an element $\beta_1 \in I_1$ such that $J_1 = \chi_{I_1}(\beta_1)$, we have $\tilde{J} \sim \beta_1\overline{I_1}I_2$. To correct this, we let $J := \beta_1^{-1}\tilde{J}\beta_1$. Then, we obtain a connecting ideal $J \sim \overline{I_1}I_2$.

We show the concrete algorithm of Δ -KLPT⁺ in Algorithm 2. In the following, we first explain `IdealNormReduce` and then discuss the randomization phase of Δ -KLPT⁺.

Algorithm 2: Δ -KLPT⁺

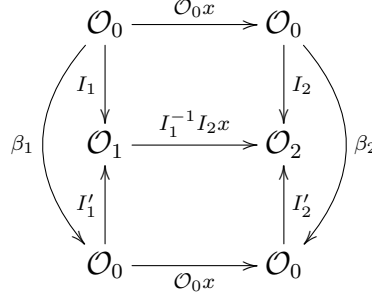
Input: left $\mathcal{O}_0$ -ideals I_1, I_2 and a prime power ℓ^e .
Output: an ideal J equivalent to $\overline{I_1}I_2$ of norm ℓ^e .
// Phase 1: Norm Alignment
1 Let e_0 be a fixed integer larger than $B_{\text{KLPT}}(p, \omega)$;
2 $L_1 \leftarrow \text{KLPT}_{2^{e_0}}(I_1)$;
3 $L_2 \leftarrow \text{KLPT}_{2^{e_0}}(I_2)$;
4 **while** *True* **do**
 // Phase 2: Randomization
5 Sample $r \xleftarrow{\$} \{0, \dots, p\}$;
6 **if** $r < p$ **then**
7 $\gamma \leftarrow 1 + ri$;
8 **else**
9 $\gamma \leftarrow i$;
10 $J_1 \leftarrow L_1 \gamma$;
11 $J_2 \leftarrow L_2 \overline{\gamma}$;
 // Phase 3: Norm Reduction
12 **while** $\text{nrd}(J_1) \geq p$ **do**
13 $(J_1, J_2) \leftarrow \text{IdealNormReduce}(J_1, J_2)$;
14 $N \leftarrow \text{nrd}(J_1)$;
15 **if** N is a prime congruent to 3 modulo 4 **then**
16 $(C : D) \leftarrow \text{IdealModConstraint}(J_1, J_2)$;
17 **if** $\left(\frac{p\ell^e(C^2 + D^2)}{N}\right) = 1$ **then**
18 break ;
 // Phase 4: Δ -KLPT
19 $\beta_1 \leftarrow$ an element in I_1 such that $J_1 = \chi_{I_1}(\beta_1)$;
20 $\alpha \leftarrow \text{FullStrongApproximation}_{\ell^e}(N, C, D)$;
21 $\tilde{J} \leftarrow J_1^{-1} \cdot J_2 \alpha$;
22 $J \leftarrow \beta_1^{-1} \tilde{J} \beta_1$;
23 **return** J ;

3.2 IdealNormReduce

In this subsection, we present the norm-reduction algorithm **IdealNormReduce** (Algorithm 3), which is the core of Δ -KLPT⁺. Let I_1 and I_2 be left $\mathcal{O}_0$ -ideals satisfying $\text{nrd}(I_1) = \text{nrd}(I_2) = N$, and let $\mathcal{O}_1$ and $\mathcal{O}_2$ be their respective right orders. The algorithm constructs equivalent ideals whose norms remain equal but are substantially smaller. We first explain the idea behind the construction, and then bound the output norm.

Rough sketch. Our starting point is the diagram in Fig. 1. We first choose a nonzero element $x \in \mathcal{O}_0$ such that the push-forward of $\mathcal{O}_0 x$ by I_1 corresponds to I_2 . For this purpose, it suffices to choose

$$x \in \mathcal{O}_0 \cap I_2^{-1} I_1.$$

**Fig. 1.** Diagram underlying `IdealNormReduce`.

Indeed, this condition is equivalent to $I_2x \subseteq I_1$. Consequently, $I_1^{-1}I_2x$ is a left $\mathcal{O}_1$ -ideal whose right order is $x^{-1}\mathcal{O}_2x \cong \mathcal{O}_2$; it therefore represents the push-forward of $\mathcal{O}_0x$ by I_1 .

We next choose a nonzero element $\beta_1 \in I_1$ such that the push-forward of the principal ideal $\mathcal{O}_0\beta_1$ by $\mathcal{O}_0x$ is again principal, say $\mathcal{O}_0\beta_2$, which is contained in I_2 . It suffices to take

$$\beta_1 \in I_1 \cap x^{-1}I_2x \quad \text{and set} \quad \beta_2 = x\beta_1x^{-1}.$$

Indeed, the intersection condition gives $\beta_2 \in I_2$, while the identity $x\beta_1 = \beta_2x$ gives the required commutative square of principal ideals.

Finally, we set $I_1' = \chi_{I_1}(\beta_1)$ and $I_2' = \chi_{I_2}(\beta_2)$. Since conjugation by x preserves the reduced norm, we have $\text{nrd}(\beta_1) = \text{nrd}(\beta_2)$ and therefore

$$\text{nrd}(I_1') = \frac{\text{nrd}(\beta_1)}{N} = \frac{\text{nrd}(\beta_2)}{N} = \text{nrd}(I_2').$$

To obtain a small output norm, we successively choose x and β_1 of minimum nonzero norm among the elements satisfying the conditions above. More precisely, we first choose a shortest nonzero element of $\mathcal{O}_0 \cap I_2^{-1}I_1$ as x , and then choose a shortest nonzero element of $I_1 \cap x^{-1}I_2x$ as β_1 . We show the concrete algorithm obtained from the above argument in Algorithm 3.

Bound for the output norm. We next show that the output norm N' obtained by `IdealNormReduce` is smaller than $\delta^{3/2}p^{3/4}N^{1/4}$. To do so, we first bound the norms of x and β_1 taken in `IdealNormReduce` using Lemma 1 and the following lattice-index bounds.

Lemma 5. *Let I_1 and I_2 be left $\mathcal{O}_0$ -ideals of the same norm N . Then, $\mathcal{O}_0 \cap I_2^{-1}I_1$ is a lattice in $\mathcal{O}_0$ and $[\mathcal{O}_0 : \mathcal{O}_0 \cap I_2^{-1}I_1] \leq N$.*

Lemma 6. *Let I_1 and I_2 be left $\mathcal{O}_0$ -ideals of the same norm N , and $x \in \mathcal{O}_0 \cap I_2^{-1}I_1$ be a nonzero element. Then, we have $[\mathcal{O}_0 : I_1 \cap x^{-1}I_2x] \leq N^2 \cdot \text{nrd}(x)$.*

Algorithm 3: IdealNormReduce**Input:** left $\mathcal{O}_0$ -ideals I_1, I_2 with common norm N .**Output:** left $\mathcal{O}_0$ -ideals I'_1, I'_2 with common norm $N' < \delta^{3/2} p^{3/4} N^{1/4}$.

- 1 $N \leftarrow \text{nrd}(I_1)$;
- 2 $L \leftarrow \mathcal{O}_0 \cap I_2^{-1} I_1$;
- 3 $x \leftarrow$ a shortest nonzero vector in L ;
- 4 $L_x \leftarrow I_1 \cap x^{-1} I_2 x$;
- 5 $\beta_1 \leftarrow$ a shortest nonzero vector in L_x ;
- 6 $\beta_2 \leftarrow x \beta_1 x^{-1}$;
- 7 **return** $\chi_{I_1}(\beta_1), \chi_{I_2}(\beta_2)$;

The proofs of these lemmas are given in Appendix C. From these lemmas, we obtain the following proposition.

Proposition 7. *The algorithm IdealNormReduce (Algorithm 3) is correct and runs in time polynomial in $\log p$ and $\log N$. In addition, there is no prime q such that both I'_1/q and I'_2/q are contained in $\mathcal{O}_0$ for the output ideals.*

Proof. By Lemma 1 and Lemma 5, a shortest nonzero vector $x \in \mathcal{O}_0 \cap I_2^{-1} I_1$ satisfies

$$\text{nrd}(x) \leq \delta \sqrt{pN}.$$

In addition, by Lemma 1 and Lemma 6, a shortest nonzero vector $\beta_1 \in I_1 \cap x^{-1} I_2 x$ satisfies

$$\text{nrd}(\beta_1) \leq \delta \sqrt{pN^2 \text{nrd}(x)} \leq \delta^{3/2} p^{3/4} N^{5/4}.$$

Therefore, the output ideals $I'_1 := \chi_{I_1}(\beta_1)$ and $I'_2 := \chi_{I_2}(\beta_2)$ have the common norm

$$N' = \frac{\text{nrd}(\beta_1)}{N} \leq \delta^{3/2} p^{3/4} N^{1/4}.$$

Since the right-hand side is transcendental, equality cannot hold. This gives the correctness of IdealNormReduce.

The two lattices used by IdealNormReduce have fixed rank four. Therefore, their shortest nonzero vectors can be computed in time polynomial in $\log p$ and $\log N$ using exact shortest-vector algorithms in fixed dimension.

Let q be a prime such that both $\chi_{I_1}(\beta_1)/q$ and $\chi_{I_2}(\beta_2)/q$ are contained in $\mathcal{O}_0$. Then, we have $\beta_1/q \in I_1$ and $\beta_2/q \in I_2$. Since $\beta_2 = x \beta_1 x^{-1}$, we have $\beta_1/q \in I_1 \cap x^{-1} I_2 x$. This contradicts the fact that β_1 is a shortest nonzero vector in $I_1 \cap x^{-1} I_2 x$. $\square$

Number of iterations. The above bound shows that whenever $N > \delta^2 p$, we have $N' < \delta^{3/2} p^{3/4} N^{1/4} < N$. Thus, each application of IdealNormReduce strictly decreases the common norm. Since the norms are integers, each such decrease is at least one. Consequently, sufficiently many iterations of IdealNormReduce yield a pair of ideals with common norm $N < \delta^2 p < p$. The following proposition bounds the number of iterations required to obtain $N < p$.

Proposition 8. *Let I_1 and I_2 be left $\mathcal{O}_0$ -ideals of the same norm $N > p$. Consider applying **IdealNormReduce** (Algorithm 3) to (I_1, I_2) iteratively until the output ideals have norm below p . Then, the number of iterations is at most*

$$\left\lceil \frac{1}{2} \log_2 \left(\log_2 \left(\frac{N}{p} \right) + C_1 \right) + C_2 \right\rceil,$$

where $C_1 = 2 \log_2(\delta^{-1}) \approx 0.303$ and $C_2 = \frac{1}{2} \log_2(C_1^{-1}) \approx 0.861$.

Proof. Let $N_0 := N$ and N_i be the norm of the output ideals after the i -th iteration of **IdealNormReduce**. Then, we have $N_{i+1} < \delta^{3/2} p^{3/4} N_i^{1/4}$. Taking logarithm base 2 of both sides and letting $x_i := \log_2(N_i/p)$, we have $x_{i+1} < \frac{1}{4}x_i - \frac{3}{2} \log_2(\delta^{-1})$. By solving this recurrence relation, we have

$$x_i < \frac{1}{4^i} \left(\log_2 \left(\frac{N}{p} \right) + 2 \log_2(\delta^{-1}) \right) - 2 \log_2(\delta^{-1}).$$

Since $N_i < p$ is equivalent to $x_i < 0$, we obtain the desired bound on the number of iterations. $\square$

Proposition 8 shows that $O(\log \log(N/p))$ iterations of **IdealNormReduce** suffice to obtain a common norm below p . By continuing the iteration, we can further reduce the common norm below $\delta^2 p$, but reaching this threshold requires $O(\log p + \log \log(N/p))$ iterations in total. In this paper, we prioritize the asymptotic running time over a constant-factor improvement in the norm bound, and therefore stop once the common norm is below p .

3.3 Effect of Randomization

All phases other than the randomization phase are performed deterministically. We therefore need to show that the initial random choice r provides enough variation in the final reduced pair. For $r \in \{0, \dots, p\}$, let $(J_{1,r}, J_{2,r})$ denote the pair of ideals obtained by iteratively applying **IdealNormReduce** after choosing r as the randomness. In this subsection, we discuss the distribution of $(J_{1,r}, J_{2,r})$ when r is chosen uniformly at random.

We first define classes of equivalent same-norm ideal pairs. Then we show that **IdealNormReduce** preserves the selected class with overwhelming probability and that the randomization phase selects one of these classes uniformly.

Invariant class. For $n \in \mathbb{Z}$, let μ_n denote the group of n -th roots of unity in $\mathbb{F}_{p^2}^\times$. To formalize the classes mentioned above, we first construct a map θ from quaternion elements to μ_{p+1} and then use it to define an invariant Ψ of pairs of equivalent same-norm ideals. Define

$$\mathcal{B}_{p,\infty}^0 := \{x \in \mathcal{B}_{p,\infty}^\times \mid v_p(\text{nrd}(x)) = 0\},$$

where v_p denotes the p -adic valuation on $\mathbb{Q}$. Thus, $\mathcal{B}_{p,\infty}^0$ is the subgroup of elements whose reduced norms are p -adic units.

Let $(\mathcal{O}_0)_p = \mathcal{O}_0 \otimes_{\mathbb{Z}} \mathbb{Z}_p$ and let $\mathfrak{P}$ denote the unique maximal ideal of $(\mathcal{O}_0)_p$. Recall that $(\mathcal{O}_0)_p/\mathfrak{P} \cong \mathbb{F}_{p^2}$ [37, Theorem 13.3.11]. This isomorphism induces a surjective homomorphism

$$\tau : \mathcal{B}_{p,\infty}^0 \longrightarrow \mathbb{F}_{p^2}^\times.$$

More concretely, if $x = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k} \in \mathcal{B}_{p,\infty}^0$, then $\tau(x) = a_0 + b_0\mathbf{i}$, where $a_0, b_0 \in \mathbb{F}_p$ are the residue classes of a, b modulo p .

We define a map $\theta : \mathcal{B}_{p,\infty}^0 \rightarrow \mu_{p+1}$; $x \mapsto \tau(x)^{1-p}$. The map θ is a group homomorphism and satisfies the following properties:

- the restriction of θ to $\mathcal{O} \cap \mathcal{B}_{p,\infty}^0$ is surjective onto μ_{p+1} for any maximal order $\mathcal{O}$ of $\mathcal{B}_{p,\infty}$;
- $\theta(x)^p = \theta(\bar{x}) = \theta(x)^{-1}$ for every $x \in \mathcal{B}_{p,\infty}^0$;
- $\theta(n) = 1$ for every $n \in \mathbb{Q} \cap \mathcal{B}_{p,\infty}^0$.

Let I_1 and I_2 be left $\mathcal{O}_0$ -ideals of the same norm prime to p , and let $\mathcal{O}_1$ and $\mathcal{O}_2$ be their respective right orders. Define

$$\mathcal{I}_{I_1, I_2}^0 := \{(J_1, J_2) \mid J_1 \sim I_1, J_2 \sim I_2, \text{nrd}(J_1) = \text{nrd}(J_2), \text{ and } p \nmid \text{nrd}(J_1)\}.$$

The following lemma associates each pair $(J_1, J_2) \in \mathcal{I}_{I_1, I_2}^0$ with a value in $\mu_{(p+1)/2}$. We use this value to classify the ideal pairs: two pairs belong to the same class if and only if their associated values agree.

Lemma 9. *The map*

$$\begin{aligned} \Psi_{I_1, I_2} : \mathcal{I}_{I_1, I_2}^0 &\longrightarrow \mu_{(p+1)/2} / \langle \theta(\mathcal{O}_1^\times), \theta(\mathcal{O}_2^\times) \rangle; \\ (J_1, J_2) &\longmapsto \theta(\beta_1)\theta(\bar{\beta}_2) \bmod \langle \theta(\mathcal{O}_1^\times), \theta(\mathcal{O}_2^\times) \rangle, \end{aligned}$$

where $\beta_1, \beta_2 \in \mathcal{B}_{p,\infty}^0$ satisfy $J_1 = I_1\beta_1$ and $J_2 = I_2\beta_2$, is well-defined and surjective.

Proof. Since $\text{nrd}(I_1) = \text{nrd}(I_2)$ and $\text{nrd}(J_1) = \text{nrd}(J_2)$ are prime to p , we have $\beta_1, \beta_2 \in \mathcal{B}_{p,\infty}^0$. Moreover, we have $\theta(\beta_1)\theta(\bar{\beta}_2) \in \mu_{(p+1)/2}$ since $\text{nrd}(\beta_1) = \text{nrd}(\beta_2)$.

Suppose that $J_1 = I_1\beta'_1$ is another representation. Then $\mathcal{O}_1\beta_1 = \mathcal{O}_1\beta'_1$, so there exists $\xi_1 \in \mathcal{O}_1^\times$ such that $\beta'_1 = \xi_1\beta_1$. Similarly, if $J_2 = I_2\beta'_2$, there exists $\xi_2 \in \mathcal{O}_2^\times$ such that $\beta'_2 = \xi_2\beta_2$. Hence $\theta(\beta'_1)\theta(\bar{\beta}'_2) \equiv \theta(\beta_1)\theta(\bar{\beta}_2) \bmod \langle \theta(\mathcal{O}_1^\times), \theta(\mathcal{O}_2^\times) \rangle$, proving well-definedness.

For surjectivity, let $\zeta \in \mu_{p+1}$. The surjectivity of θ on $\mathcal{O}_0 \cap \mathcal{B}_{p,\infty}^0$ gives an $\alpha \in \mathcal{O}_0 \cap \mathcal{B}_{p,\infty}^0$ such that $\theta(\alpha) = \zeta$. Setting $J_1 = I_1\alpha$ and $J_2 = I_2\bar{\alpha}$ gives $\text{nrd}(J_1) = \text{nrd}(J_2)$ and $\Psi_{I_1, I_2}(J_1, J_2) = \zeta^2$. Since the squaring map from μ_{p+1} onto $\mu_{(p+1)/2}$ is surjective, the result follows. $\square$

Remark 10. For a generic maximal order $\mathcal{O}$, we have $\mathcal{O}^\times = \{\pm 1\}$; the only exceptions are the maximal orders corresponding to elliptic curves with $j = 0$ or 1728. Since $\theta(\pm 1) = 1$, it follows that $\theta(\mathcal{O}^\times) = \{1\}$. Thus, when both $\mathcal{O}_1$ and $\mathcal{O}_2$ are generic, the codomain of Ψ_{I_1, I_2} is simply $\mu_{(p+1)/2}$.

Preservation by IdealNormReduce. We next show that applying `IdealNormReduce` does not change the invariant class of an ideal pair with overwhelming probability.

Proposition 11. *Let I_1 and I_2 be left $\mathcal{O}_0$ -ideals of norm prime to p and let $(J_1, J_2) \in \mathcal{I}_{I_1, I_2}^0$. Let x, β_1, β_2 be the elements computed by `IdealNormReduce` with input J_1 and J_2 , and let $J'_1 = \chi_{J_1}(\beta_1)$ and $J'_2 = \chi_{J_2}(\beta_2)$ be its output ideals. If $\text{nrd}(x)$ and $\text{nrd}(\beta_1)$ are prime to p , then (J'_1, J'_2) belongs to the same invariant class as (J_1, J_2) ; equivalently, $\Psi_{I_1, I_2}(J'_1, J'_2) = \Psi_{I_1, I_2}(J_1, J_2)$.*

Proof. The assumptions imply that x, β_1 , and β_2 belong to $\mathcal{B}_{p, \infty}^0$. Since $\beta_2 = x\beta_1x^{-1}$ and θ is a homomorphism, we have $\theta(\beta_2) = \theta(\beta_1)$. Moreover, $\theta(\bar{\beta}_2) = \theta(\beta_2)^{-1}$. Therefore, $\Psi_{I_1, I_2}(J'_1, J'_2) = \Psi_{I_1, I_2}(J_1, J_2)\theta(\beta_1)\theta(\bar{\beta}_2) = \Psi_{I_1, I_2}(J_1, J_2)$, which proves the claim. $\square$

We use the following heuristic to argue that $\text{nrd}(x)$ and $\text{nrd}(\beta_1)$ are prime to p with overwhelming probability.

Heuristic 12 *Let N be a positive integer prime to p , and sample left $\mathcal{O}_0$ -ideals J_1 and J_2 uniformly at random from the set of left $\mathcal{O}_0$ -ideals of norm N such that there is no prime q for which both J_1/q and J_2/q are contained in $\mathcal{O}_0$. Then, the probability that p divides $\text{nrd}(x)$ in `IdealNormReduce`(J_1, J_2) is in $O(1/p)$. Conditioned on $p \nmid \text{nrd}(x)$, the probability that p divides $\text{nrd}(\beta_1)$ is in $O(1/p^2)$.*

An explanation of this heuristic is given in Appendix B.2.

Uniformity of the randomization. Let I_1 and I_2 be left $\mathcal{O}_0$ -ideals of the same norm prime to p . Their right-order unit groups are $\{\pm 1\}$ with overwhelming probability, so by Remark 10, the invariant takes values in $\mu_{(p+1)/2}$. Let $S := \{1 + r\mathbf{i} \mid r \in \{0, 1, \dots, p-1\}\} \cup \{\mathbf{i}\}$. For $\gamma \in S$, the randomization phase sets $J_1 = I_1\gamma$ and $J_2 = I_2\bar{\gamma}$, and therefore $\Psi_{I_1, I_2}(J_1, J_2) = \theta(\gamma^2)$. The elements of S represent the $p+1$ classes of $\mathbb{F}_{p^2}^\times / \mathbb{F}_p^\times$. Since the kernel of $z \mapsto z^{1-p}$ on $\mathbb{F}_{p^2}^\times$ is $\mathbb{F}_p^\times$, the map $\gamma \mapsto \theta(\gamma)$ is a bijection from S to μ_{p+1} . The squaring map from μ_{p+1} to $\mu_{(p+1)/2}$ is two-to-one. Hence $\gamma \mapsto \theta(\gamma^2)$ is two-to-one onto $\mu_{(p+1)/2}$, and a uniformly sampled γ produces a uniformly random invariant class.

We assume that all routines other than the choice of γ are deterministic. Under Heuristic 12, each application of `IdealNormReduce` preserves the selected invariant class with overwhelming probability. Consequently, the final pair obtained after the norm-reduction phase is uniformly distributed over the outcomes associated with the $(p+1)/2$ invariant classes.

Remark 13. We expect only a constant fraction of the distinct ideal pairs (J_1, J_2) with common norm $N < p$ to have the same common norms. Indeed, for a fixed $N < p$, the number of ideals of norm N in a given left $\mathcal{O}_0$ -ideal class is heuristically small, of order $O(N/p) = O(1)$. In particular, distinct pairs are expected to have distinct first components J_1 .

We further expect distinct ideal pairs obtained after the while loop in Algorithm 2 to give distinct outputs of $\Delta\text{-KLPT}^+$. Suppose, for contradiction, that

two distinct pairs yield the same output ideal J . By the properties of Δ -KLPT, this would imply the existence of two distinct ideals J_1 of prime norm less than p such that pulling back J along J_1 yields a principal $\mathcal{O}_0$ -ideal. Heuristically, the number of ideals of prime norm less than p in a fixed ideal class is $O(p/\log p)$. Among these, only a proportion of $O(1/p)$ is expected to satisfy the pullback condition, leaving only $O(1/\log p)$ such ideals. Hence, the occurrence of two distinct J_1 satisfying this condition is expected to be unlikely.

3.4 Analysis of Δ -KLPT⁺

In this subsection, we give an analysis of the target norm, success probability, and running time of Δ -KLPT⁺ (Theorem 15). Our analysis uses the following heuristic assumptions.

Heuristic 14 *In Algorithm 2, for each $r \in \{0, \dots, p\}$, let $(J_{1,r}, J_{2,r})$ be the pair of ideals ultimately obtained by the norm-reduction phase, and let $N_r < p$ be their common norm. Then*

$$\Pr_{r \xleftarrow{\$} \{0, \dots, p\}} [(J_{1,r}, J_{2,r}) \text{ satisfies Conditions 2 and 3 in (1)}] \approx \frac{1}{4 \log p}.$$

Explanation. By Heuristic 12, the applications of `IdealNormReduce` preserve the invariant class selected by the randomization phase with overwhelming probability. Thus, the ideal pair $(J_{1,r}, J_{2,r})$ ranges over approximately $(p+1)/2$ distinct ideal pairs with overwhelming probability.

We further expect a constant fraction of the invariant classes to have pairwise distinct common norms as discussed in Remark 13. We therefore expect the values N_r also to take $\Theta(p)$ distinct values. Thus, the values N_r cover a substantial fraction of the positive integers below p . Assuming that there is no structural bias toward a particular subset of these integers, it is therefore natural to model N_r as an approximately uniform positive integer smaller than p . By the prime number theorem for arithmetic progressions, the probability that N_r is prime congruent to 3 modulo 4 is approximately $1/(2 \log p)$.

Furthermore, assume that, conditioned on N_r being prime congruent to 3 modulo 4, $\text{QR}_{p, \ell^e}(J_{1,r}, J_{2,r})$ is uniformly distributed over $\{1, -1\}$. Condition $\text{QR}_{p, \ell^e}(J_{1,r}, J_{2,r}) = 1$ then holds with probability $1/2$. This gives the claimed probability $1/(4 \log p)$.

Now we give an analysis of Δ -KLPT⁺. Proposition 7 establishes the correctness and polynomial-time complexity of each norm-reduction step, while Proposition 8 bounds the number of such steps. Combining these results with the three heuristic estimates above yields the following theorem.

Theorem 15. *Assume Heuristics 2, 3, and 14. Let $\omega > 0$ and $\omega' := \omega + 2$. Choose an integer e_0 with $e_0 > B_{\text{KLPT}}(p, \omega')$. If the prime power ℓ^e satisfies*

$$\ell^e > \frac{6 \log(2)}{\pi} \omega' p^4 \log p$$

then $\Delta\text{-KLPT}^+_{\ell^e}$ is correct, runs in expected time polynomial in $\log p$ and ω , and succeeds with probability at least $1 - 2^{-\omega}$.

Proof. Correctness follows from the construction described in Subsection 3.1. It remains to prove the expected running time and the success probability.

We analyze the running time. By Proposition 7, each call to **IdealNormReduce** runs in time polynomial in $\log p$. Proposition 8 and Heuristic 14 imply that the expected total number of **IdealNormReduce** calls is $O(\log p \cdot \log(\log p))$. Moreover, Heuristics 3 and 2 ensure that **KLPT** and **FullStrongApproximation** run in expected time polynomial in $\log p$ and ω . Therefore, the expected running time of $\Delta\text{-KLPT}^+_{\ell^e}$ is polynomial in $\log p$ and ω .

Finally, we analyze the success probability. By Heuristic 3, each of the two calls to **KLPT** in the norm-alignment phase fails with probability at most $2^{-\omega'}$. Let $N < p$ be the common prime norm of the resulting pair. The norm bound assumption gives

$$\ell^e > \frac{3 \log(2)}{2\pi} \omega' p N^3 \log(p N^3),$$

since $p N^3 < p^4$ and $\log(p N^3) < 4 \log p$. According to Heuristic 2, the failure probability of **FullStrongApproximation** is at most $2^{-\omega'}$. A union bound over the two **KLPT** calls and the **FullStrongApproximation** call gives total failure probability at most $3 \cdot 2^{-\omega'} = \frac{3}{4} 2^{-\omega} < 2^{-\omega}$. $\square$

Furthermore, we discuss some practical optimizations for reducing the number of **IdealNormReduce** calls in Appendix D.

3.5 $\Delta\text{-SigningKLPT}^+$

For the application to signatures, the output $J \leftarrow \Delta\text{-KLPT}^+_{\ell^e}(I_1, I_2)$ must satisfy the additional condition $I_1 J \not\subseteq \ell\mathcal{O}_0$. In our Σ -protocol in Section 4, this condition means that the composition $\varphi_{\text{rsp}} \circ \varphi_{\text{chl}}$ of the response and challenge isogenies is cyclic and is essential for special soundness. The generic version of $\Delta\text{-KLPT}^+$ analyzed above does not impose this condition. We refer to the variant satisfying this condition as $\Delta\text{-SigningKLPT}^+$. Note that we require its first input ideal I_1 to be primitive.

We first express the condition on $I_1 J$ in terms of the element α produced by the final **FullStrongApproximation** call in line 20.

Lemma 16. *Let I_1 and I_2 be left $\mathcal{O}_0$ -ideals, and set $J \leftarrow \Delta\text{-KLPT}^+_{\ell^e}(I_1, I_2)$. Let J_1 and J_2 be the outputs of the final **IdealNormReduce** call in this execution. Let $N = \text{nrd}(J_1) = \text{nrd}(J_2) \neq \ell$, and let $\beta_1 \in I_1$ be such that $J_1 = \chi_{I_1}(\beta_1)$. Let α be the output of the final **FullStrongApproximation** call. Then*

$$I_1 J \not\subseteq \ell\mathcal{O}_0 \iff \alpha\beta_1 \notin \ell\mathcal{O}_0.$$

Proof. From the construction of $\Delta\text{-KLPT}^+$, $J = \beta_1^{-1} J_1^{-1} J_2 \alpha \beta_1 = \frac{1}{N} I_1^{-1} J_2 \alpha \beta_1$. Set $K := I_1 J$ and $x := \alpha \beta_1$. Then we have $K = N^{-1} J_2 x$. Since $N \in J_2$, we have $x \in K$. Therefore, $K \subseteq \ell\mathcal{O}_0$ implies $x \in \ell\mathcal{O}_0$.

Conversely, suppose that $x \in \ell\mathcal{O}_0$ and write $x = \ell y$ with $y \in \mathcal{O}_0$. Since $K = (\ell/N)J_2y \subseteq \mathcal{O}_0$, we have $\ell J_2y = NK \subseteq N\mathcal{O}_0$. Since ℓ is invertible modulo N , this implies $J_2y \subseteq N\mathcal{O}_0$, and therefore $K = \ell \left(\frac{1}{N}J_2y\right) \subseteq \ell\mathcal{O}_0$. $\square$

We therefore modify `FullStrongApproximation` to take β_1 as an additional input, writing $\alpha \leftarrow \text{FullStrongApproximation}_{\ell^e}(N, C, D, \beta_1)$. This modified procedure accepts a candidate α only if $\alpha\beta_1 \notin \ell\mathcal{O}_0$. By Lemma 16, this test is equivalent to the required condition $I_1J \not\subseteq \ell\mathcal{O}_0$. We give the concrete algorithm for this modified `FullStrongApproximation` in Algorithm 4 in Appendix B.

We use the following signing analogue of Heuristic 2 to estimate the effect of this additional condition.

Heuristic 17 *Let N and ℓ be primes such that $N \neq p, \ell$, and suppose that ℓ^e, N, C, D satisfy $\left(\frac{p\ell^e(C^2+D^2)}{N}\right) = 1$. Let $\beta_1 \in \mathcal{O}_0$ such that $\beta_1 \notin \ell\mathcal{O}_0$. If*

$$\ell^e > \frac{3 \log(2)}{2\pi} \frac{\ell^2 + \ell - 1}{\ell^2 - 1} \omega \cdot pN^3 \log(pN^3),$$

then the modified algorithm `FullStrongApproximation` $_{\ell^e}(N, C, D, \beta_1)$ returns an element α such that $\alpha\beta_1 \notin \ell\mathcal{O}_0$ with probability at least $1 - 2^{-\omega}$, and runs in expected time polynomial in $\log p$, $\log N$, and ω .

An explanation of this heuristic is given in Appendix B. Finally, replacing Heuristic 2 with Heuristic 17 in the analysis of Theorem 15 gives the following analogous result.

Theorem 18. *Assume Heuristics 3, 14, and 17. Let $\omega > 0$ and $\omega' := \omega + 2$. Choose an integer e_0 with $e_0 > B_{\text{KLPT}}(p, \omega')$. If the target prime power ℓ^e satisfies*

$$\ell^e > \frac{6 \log(2)}{\pi} \frac{\ell^2 + \ell - 1}{\ell^2 - 1} \omega' p^4 \log p,$$

then $\Delta\text{-SigningKLPT}^+_{\ell^e}$ is correct, runs in expected time polynomial in $\log p$ and ω , and succeeds with probability at least $1 - 2^{-\omega}$.

4 Application: Ring Signature Scheme Delfar

We now demonstrate the cryptographic impact of the improved $\tilde{O}(p^4)$ bound by constructing the fully anonymous ring signature scheme Delfar.

The overall design of Delfar follows that of Erebor-full [11]. We first construct a Σ -protocol using the same diagram as Erebor-full, shown below, and then apply the AOS framework [4] to obtain a ring signature scheme. For ease of exposition, we reverse the direction of the response isogeny φ_{rsp} relative to Erebor-full; this is not an essential difference.

$$\begin{array}{ccc} E_0 & \xrightarrow{\varphi_{\text{sk}}} & E_{\text{pk}} \\ \varphi_{\text{com}} \downarrow & & \downarrow \varphi_{\text{chl}} \\ E_{\text{com}} & \xleftarrow{\varphi_{\text{rsp}}} & E_{\text{chl}} \end{array}$$

The main difference from Erebor-full lies in the algorithm used to generate the ideal corresponding to the response isogeny φ_{rsp} . Erebor-full uses a modified version of **SigningKLPT**, whereas Delfar uses Δ -**SigningKLPT**⁺. The smaller norm bound of Δ -**SigningKLPT**⁺ allows the response isogeny to have a smaller degree.

Parameters. In our Σ -protocol, we use the following parameters for the security parameter λ :

- A large prime $p = c \cdot 2^f - 1$ with $f > 2\lambda$ and $c \in \mathbb{N}$ as small as possible.⁷ By the prime number theorem for arithmetic progressions, the smallest such c is heuristically expected to be $O(\lambda)$.
- An elliptic curve $E_0 : y^2 = x^3 + x$ over $\mathbb{F}_{p^2}$.
- A maximal order $\mathcal{O}_0 = \mathbb{Z}\langle 1, \mathbf{i}, \frac{1+\mathbf{j}}{2}, \frac{1+\mathbf{k}}{2} \rangle$, which is isomorphic to $\text{End}(E_0)$.
- An integer $D_{\text{rsp}} = 2^{e_{\text{rsp}}} = 2^{4f+\epsilon}$, where ϵ is chosen so that

$$2^{4f+\epsilon} > \frac{10 \log(2)}{\pi} (\lambda + 2) p^4 \log p.$$

This inequality is obtained from Theorem 18 by setting $\ell = 2$ and $\omega = \lambda$, and ensures that Δ -**SigningKLPT**⁺ _{D_{rsp}} succeeds with probability at least $1 - 2^{-\lambda}$. Solving the inequality for ϵ , using $p < c \cdot 2^f$ and $c = O(\lambda)$, shows that $\epsilon = O(\log \lambda)$ suffices.

- A prime $D_{\text{mix}} > p \cdot 2^{2\lambda}$, which is used as the degree of φ_{sk} and φ_{com} .

We use the same p and D_{mix} as those of NIST SQIsign in Round 3 [1]. Under the setting, it holds that $\epsilon + \lambda < f$. We use this property in the response generation.

4.1 Underlying Σ -Protocol

In the following, we describe an overview of our Σ -protocol.

- **Key generation.** Compute a random D_{mix} -isogeny $\varphi_{\text{sk}} : E_0 \rightarrow E_{\text{pk}}$ and its corresponding ideal I_{sk} . Set $\text{pk} = E_{\text{pk}}$ and $\text{sk} = (\varphi_{\text{sk}}, I_{\text{sk}})$.
- **Commitment.** Compute a random D_{mix} -isogeny $\varphi_{\text{com}} : E_0 \rightarrow E_{\text{com}}$ and its corresponding ideal I_{com} . Send E_{com} as a commitment.
- **Challenge.** We use $\mathcal{C} := \mathbb{Z}/2^\lambda\mathbb{Z}$ as the challenge space. Given a challenge $\text{chl} \xleftarrow{\$} \mathcal{C}$, we define a 2^λ -isogeny $\varphi_{\text{chl}} : E_{\text{pk}} \rightarrow E_{\text{chl}}$ by $\ker(\varphi_{\text{chl}}) = \langle P_{\text{pk}} + [\text{chl}]Q_{\text{pk}} \rangle$, where $(P_{\text{pk}}, Q_{\text{pk}})$ is a deterministic basis of $E_{\text{pk}}[2^\lambda]$.
- **Response.** Let I_{chl} be the ideal corresponding to φ_{chl} and let $I_1 = I_{\text{sk}}I_{\text{chl}}$ and $I_2 = I_{\text{com}}$. Compute $J_{\text{rsp}} \leftarrow \Delta$ -**SigningKLPT**⁺ _{D_{rsp}} (I_1, I_2) and return its corresponding isogeny $\varphi_{\text{rsp}} : E_{\text{chl}} \rightarrow E_{\text{com}}$ as a response.

⁷ Until recently, choosing $f \approx 2\lambda$ was considered sufficient. Wesolowski recently proposed an attack on the supersingular isogeny problem requiring time and memory $p^{1/3+o(1)}$ [39], suggesting that larger values of p may now be necessary. For example, SQIsign in Round 3 uses $f \approx 2\lambda + 80, 2\lambda + 120, 2\lambda + 160$ for NIST security level I, III, V, respectively [1].

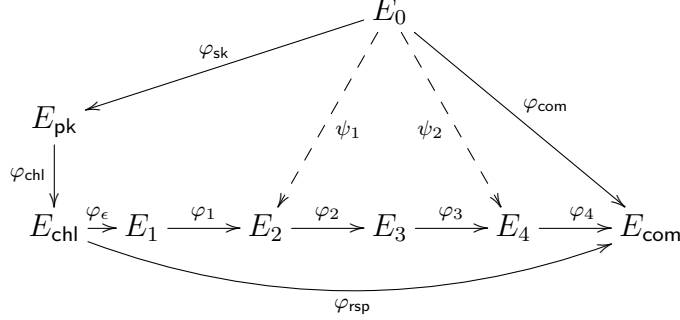


Fig. 2. Overview of the computation of the response isogeny.

- **Verification.** Check that φ_{rsp} is an isogeny from E_{chl} to E_{com} and that $\varphi_{\text{rsp}} \circ \varphi_{\text{chl}}$ is cyclic.

Note that our key generation, commitment, and challenge are the same as those of NIST SQIsign in Round 3 [1].

We now describe in detail how to compute the response isogeny φ_{rsp} corresponding to J_{rsp} . We factor J_{rsp} as $J_{\text{rsp}} = J_{\epsilon} J_1 J_2 J_3 J_4$, where $\text{nrd}(J_{\epsilon}) = 2^{\epsilon}$ and $\text{nrd}(J_i) = 2^f$ for $i \in \{1, 2, 3, 4\}$. Let $\varphi_{\epsilon}, \varphi_1, \varphi_2, \varphi_3, \varphi_4$ denote the isogenies corresponding to $J_{\epsilon}, J_1, J_2, J_3, J_4$, respectively. For clarity, we illustrate the overall procedure in Fig. 2. Our goal is to compute generators of $\ker(\varphi_{\epsilon})$ and $\ker(\varphi_i)$ for $i \in \{1, 2, 3, 4\}$.

We first compute a generator K_{ϵ} of $\ker(\varphi_{\epsilon})$ as follows. We compute a generator of $E_0[I_1 J_{\epsilon}] \cap E_0[2^{\lambda+\epsilon}]$ and evaluate it under $\varphi_{\text{chl}} \circ \varphi_{\text{sk}}$. Then the resulting point K_{ϵ} generates $\ker(\varphi_{\epsilon})$. Since $\epsilon + \lambda < f$, we have $E_0[2^{\lambda+\epsilon}] \subseteq E_0[2^f] \subseteq E_0(\mathbb{F}_{p^2})$. Therefore, the computation of K_{ϵ} can be performed entirely over $\mathbb{F}_{p^2}$.

We next compute generators of $\ker(\varphi_i)$ for $i \in \{1, 2, 3, 4\}$. To this end, we choose a left $\mathcal{O}_0$ -ideal $\widetilde{L}_1 = \chi_{I_1 J_{\epsilon} J_1}(\gamma_1)$ of odd norm equivalent to $I_1 J_{\epsilon} J_1$, and a left $\mathcal{O}_0$ -ideal $\widetilde{L}_2 = \chi_{I_2 J_4}(\gamma_2)$ of odd norm equivalent to $I_2 J_4$. We then set $L_1 = \gamma_1^{-1} \widetilde{L}_1 \gamma_1$ and $L_2 = \gamma_2^{-1} \widetilde{L}_2 \gamma_2$ so that $\mathcal{O}_R(L_1) = \mathcal{O}_L(\overline{J_1})$ and $\mathcal{O}_R(L_2) = \mathcal{O}_L(\overline{J_4})$. Via **IdealToIsogeny**, we compute the isogenies ψ_1 and ψ_2 corresponding to $\widetilde{L}_1$ and $\widetilde{L}_2$, respectively.

Using ψ_1 and ψ_2 , and proceeding similarly to the computation of K_{ϵ} , we obtain generators K_1, K_2, K_3, K_4 of $\ker(\varphi_1), \ker(\varphi_2), \ker(\varphi_3), \ker(\varphi_4)$, respectively. For example, by finding a generator of $E_0[L_1 \overline{J_1}] \cap E_0[2^f]$ and evaluating it under ψ_1 , we obtain a generator K'_1 of $\ker(\widehat{\varphi}_1)$. We can easily compute a generator K_1 of $\ker(\varphi_1)$ from a generator K'_1 of $\ker(\widehat{\varphi}_1)$. For example, see [25].

Finally, we return $\text{rsp} = (K_{\epsilon}, K_1, K_2, K_3, K_4)$, from which we can compute φ_{rsp} using Vélu's formulas.

4.2 Special HVZK Simulator

To apply the AOS transform, we need to construct a special HVZK simulator $\mathcal{S}$ that, given a public key $\mathbf{pk}$ and a challenge $\mathbf{chl}$, outputs a commitment-response pair $(\mathbf{com}, \mathbf{rsp})$ forming a valid transcript $(\mathbf{com}, \mathbf{chl}, \mathbf{rsp})$. We consider the following simulator.

- $\mathcal{S}(\mathbf{pk} := E_{\mathbf{pk}}, \mathbf{chl}) \rightarrow (\mathbf{com}, \mathbf{rsp})$:
 1. Let $(P_{\mathbf{pk}}, Q_{\mathbf{pk}})$ be a deterministic basis of $E_{\mathbf{pk}}[2^\lambda]$.
 2. Compute the 2^λ -isogeny $\varphi_{\mathbf{chl}} : E_{\mathbf{pk}} \rightarrow E_{\mathbf{chl}}$ such that $\ker(\varphi_{\mathbf{chl}}) = \langle P_{\mathbf{pk}} + [\mathbf{chl}]Q_{\mathbf{pk}} \rangle$.
 3. Sample uniformly at random a cyclic $D_{\mathbf{rsp}}$ -isogeny $\varphi_{\mathbf{rsp}} : E_{\mathbf{chl}} \rightarrow E_{\mathbf{com}}$ such that $\varphi_{\mathbf{rsp}} \circ \varphi_{\mathbf{chl}}$ is cyclic.
 4. Let $\mathbf{com} = E_{\mathbf{com}}$ and $\mathbf{rsp} = (K_\epsilon, K_1, \dots, K_4)$, which are torsion points associated with the isogeny $\varphi_{\mathbf{rsp}}$ as described in Subsection 4.1.

By applying the AOS transform to our Σ -protocol together with the simulator $\mathcal{S}$ above, we obtain the ring signature scheme Delfar. The validity of this simulator is discussed in Subsection 4.3.

4.3 Security Analysis

We analyze the security of Delfar through the security of its underlying Σ -protocol. The correctness of our Σ -protocol and its special soundness with respect to the one-endomorphism problem [29] follow in the same manner as in Erebor. Therefore, it remains only to establish special strong HVZK.

To establish special strong HVZK, we require an assumption on the hardness of the following distinguishing problem.

Problem 19. Let $\mathcal{O}_0, D_{\mathbf{rsp}}, D_{\mathbf{mix}}$ be as defined in our parameter settings. Given a primitive left $\mathcal{O}_0$ -ideal I_1 of norm $2^\lambda D_{\mathbf{mix}}$ and a primitive left $\mathcal{O}_R(I_1)$ -ideal J of norm $D_{\mathbf{rsp}}$, distinguish from which of the two distributions J has been sampled:

1. $J \leftarrow \Delta\text{-SigningKLPT}^+_{D_{\mathbf{rsp}}}(I_1, I_2)$, where I_2 is a uniformly chosen left $\mathcal{O}_0$ -ideal of norm $D_{\mathbf{mix}}$.
2. J is uniformly chosen among all primitive left $\mathcal{O}_R(I_1)$ -ideals of norm $D_{\mathbf{rsp}}$ such that $I_1 J$ is primitive.

We next show that the hardness of this problem suffices to establish special strong HVZK.

Proposition 20. *The special strong HVZK of our Σ -protocol reduces to the hardness of Problem 19.*

Proof. Special strong HVZK requires that honest and simulated transcripts remain computationally indistinguishable even when the secret randomness used in key generation is revealed. In our protocol, the secret information generated

during key generation is essentially the left $\mathcal{O}_0$ -ideal I_{sk} . The challenge chl is uniform in $\mathcal{C}$ in both experiments. It therefore suffices to compare the distributions of (com, rsp) given I_{sk} and chl .

Giving $(I_{\text{sk}}, \text{chl})$ is equivalent to giving I_1 . Write $\mathcal{O}_1 = \mathcal{O}_R(I_1)$. In addition, giving the commitment-response pair (com, rsp) is equivalent to giving a primitive left $\mathcal{O}_1$ -ideal J of norm D_{rsp} . Writing $\mathcal{O}_2 = \mathcal{O}_R(J)$, the commitment curve E_{com} corresponds to $\mathcal{O}_2$, and the response isogeny $\varphi_{\text{rsp}} : E_{\text{chl}} \rightarrow E_{\text{com}}$ corresponds to J . Therefore, distinguishing the two transcript distributions reduces to distinguishing the corresponding distributions of J given I_1 .

In an honest execution, a uniformly random left $\mathcal{O}_0$ -ideal I_2 of norm D_{mix} is generated, and the response ideal is $J \leftarrow \Delta\text{-SigningKLPT}^+_{D_{\text{rsp}}}(I_1, I_2)$. In contrast, the simulator in Subsection 4.2 samples J uniformly among all primitive left $\mathcal{O}_1$ -ideals of norm D_{rsp} such that $I_1 J$ is primitive. These are precisely the two distributions in Problem 19. Its hardness therefore implies special strong HVZK. $\square$

Consequently, by Proposition 4, we obtain the following theorem.

Theorem 21. *Assume that the one-endomorphism problem is hard for Delfar public keys and that Problem 19 is hard. Then, in the random oracle model, Delfar is correct, unforgeable with respect to insider corruption, and fully anonymous.*

Hardness analysis. We now discuss the hardness of Problem 19. We first discuss the distribution of the isomorphism class of the right order $\mathcal{O}_2$ of J , which corresponds to E_{com} . The honest commitment is the endpoint of a random D_{mix} -isogeny from E_0 , while the simulated commitment is the endpoint of a cyclic D_{rsp} -isogeny from E_{chl} . Both are expected to be close to the stationary distribution. For the honest commitment, the distance from this distribution is $O(\sqrt{p/D_{\text{mix}}})$ [14, Proposition 1.12], which is negligible for $D_{\text{mix}} > p \cdot 2^{2\lambda}$. For the simulated commitment, the distance is $O(\log D_{\text{rsp}} \sqrt{p/D_{\text{rsp}}})$ [14, Proposition 1.12], which is also negligible. A distinguisher must therefore exploit the conditional distribution of J provided I_1 and the isomorphism class of $\mathcal{O}_2$.

From now on, we consider an easier distinguishing problem: instead of merely giving the isomorphism class of $\mathcal{O}_2$, we fix and reveal the connecting ideal I_2 from $\mathcal{O}_0$ to $\mathcal{O}_2$ of norm D_{mix} used as input to $\Delta\text{-SigningKLPT}^+$. Thus, given connecting ideals I_1 and I_2 from $\mathcal{O}_0$ to $\mathcal{O}_1$ and $\mathcal{O}_2$, respectively, the task is to distinguish whether J is sampled as $J \leftarrow \Delta\text{-SigningKLPT}^+_{D_{\text{rsp}}}(I_1, I_2)$ or uniformly from the set of ideals of norm D_{rsp} equivalent to $\bar{I}_1 I_2$ and for which $I_1 J$ is primitive. We denote the former distribution by $\mathcal{D}_1$ and the latter by $\mathcal{D}_2$. In the following, we present two possible attacks against this distinguishing problem.

The first is a natural resampling attack: run $\Delta\text{-SigningKLPT}^+_{D_{\text{rsp}}}(I_1, I_2)$ to obtain an output J' and compare it with J , declaring $\mathcal{D}_1$ if they coincide. The output J' is uniquely determined by the internal pair (J_1, J_2) , which is the output of the final `IdealNormReduce` call. From the discussion in Subsection 3.3 and Heuristic 14, we can assume that the number of such pairs is $M = \Theta(p/\log p)$.

From the discussion in Remark 13, we expect distinct pairs to produce distinct output ideals J' . Therefore the output distribution is close to uniform over M distinct ideals as well. Thus, for $J \xleftarrow{\$} \mathcal{D}_1$, the probability that at least one of $m = \text{poly}(\lambda)$ independent executions produces J is $1 - (1 - 1/M)^m \approx m/M = \Theta(m \log p/p)$, which is negligible. Hence, this resampling attack has negligible distinguishing advantage.

A more targeted strategy is to search directly for an ideal J_1 satisfying a constraint imposed by $\Delta\text{-SigningKLPT}^+$. Namely, if the response ideal J is sampled from $\mathcal{D}_1$, there exists an ideal $J_1 \sim I_1$ of prime norm less than p such that the pull-back $[J_1]^*J$ is a principal left $\mathcal{O}_0$ -ideal of norm D_{rsp} . If J is sampled from $\mathcal{D}_2$, we estimate that the probability that such a J_1 exists is at most $O(1/\log p)$ from the discussion in Remark 13. Thus, efficiently finding such a J_1 would give a distinguisher. Finding such J_1 amounts to solving the pull-back equation $[J_1]^*J = \mathcal{O}_0\alpha$ for unknown α together with the requirements that $J_1 \sim I_1$ and has an unknown prime norm below p . However, we do not know how to enforce these requirements simultaneously in polynomial time. Searching over α does not obviously help either, since there are exponentially many elements of norm D_{rsp} . Thus, no method is currently known for efficiently recovering J_1 from the principality constraint.

4.4 Efficiency Analysis

In this subsection, we analyze the efficiency of Delfar. In this analysis, we use the parameters specified for NIST SQIsign in Round 3 [1, Section 5.2]. For comparison with Erebor, we also use the Round 2 parameters from [2, Section 5.2]. We show the parameters in Table 1.

Table 1. Parameters of Delfar and comparison of signature sizes of Delfar and Erebor-full. The signature size for Erebor-full is based on the parameter p_{3923} from [18].

λ	Parameters of Delfar			Signature sizes (bytes)	
	$p = c \cdot 2^f - 1$	e_{rsp}	ϵ	Delfar	Erebor-full
128	$5 \cdot 2^{248} - 1$	1017	25	$16 + n \cdot 130$	$16 + n \cdot 181$
128	$3 \cdot 2^{324} - 1$	1319	23	$16 + n \cdot 167$	-
192	$27 \cdot 2^{500} - 1$	2037	37	$24 + n \cdot 257$	-
256	$17 \cdot 2^{664} - 1$	2691	35	$32 + n \cdot 339$	-

Signature sizes. The signature σ of Delfar for a ring of n parties initially has the form $\sigma = (\text{chl}_1, \{\text{com}_j\}_{j \in [n]}, \{\text{rsp}_j\}_{j \in [n]})$, where each com_j is a curve $E_{\text{com},j}$ and each rsp_j represents a $2^{e_{\text{rsp}}}$ -isogeny $\varphi_{\text{rsp},j} : E_{\text{chl},j} \rightarrow E_{\text{com},j}$ for $j \in [n]$. Since $E_{\text{com},j}$ is the codomain of $\varphi_{\text{rsp},j}$, it can be recovered from rsp_j . Hence, the curves

$\{E_{\text{com},j}\}_{j \in [n]}$ do not need to be explicitly included in the signature, and σ can be represented as $\sigma = (\text{chl}_1, \text{rsp}_1, \dots, \text{rsp}_n)$.

As described in [17, Section 8.6], each response rsp_j , representing a $2^{e_{\text{rsp}}}$ -isogeny, can be compressed into $e_{\text{rsp}} + 4(\lceil e_{\text{rsp}}/f \rceil - 1)$ bits. Since chl_1 is represented by λ bits and $e_{\text{rsp}} = 4f + \epsilon$, the total signature size is

$$\lambda + n(e_{\text{rsp}} + 4(\lceil e_{\text{rsp}}/f \rceil - 1)) = \lambda + n(4f + \epsilon + 16) \approx 4n \log_2 p + \lambda \text{ bits},$$

where the approximation assumes that ϵ is sufficiently small compared with f and $p \approx 2^f$. For comparison, the signature size of Erebor-full is approximately $(\frac{9}{2} \log_2 p + \frac{3}{2} \lambda) n + \lambda$ bits since $e_{\text{rsp}} \approx \frac{9}{2} \log_2 p + \frac{3}{2} \lambda$. Therefore, Delfar reduces the signature size by approximately $(\frac{1}{2} \log_2 p + \frac{3}{2} \lambda) n$ bits. Table 1 shows the signature sizes of Delfar and Erebor-full. We note that, as in the original Erebor paper [11], the signature size reported for Erebor-full is based on the parameter p_{3923} from [18].

Computational cost. By construction of the AOS transform, signing requires one commitment generation, one response generation, and $n - 1$ executions of the simulator $\mathcal{S}$ for a ring of n parties. Therefore, when n is sufficiently large, the signing cost is dominated by the executions of $\mathcal{S}$.

In both Erebor-full and Delfar, each execution of $\mathcal{S}$ requires the computation of a challenge isogeny φ_{chl} and a simulated response isogeny φ_{rsp} . The main difference lies in the degree of φ_{rsp} : it is approximately $p^{9/2} \cdot 2^{3\lambda/2}$ in Erebor-full, whereas it is approximately p^4 in Delfar. Since the degree of φ_{chl} is 2^λ in both cases, Delfar reduces the dominant signing cost compared to Erebor-full.

Verification essentially consists of performing the verification procedure of the underlying Σ -protocol for each of the n ring members. For both Erebor-full and Delfar, each verification requires the computation of a challenge isogeny φ_{chl} and a response isogeny φ_{rsp} again. Hence, Delfar reduces the dominant verification cost compared to Erebor-full.

4.5 Implementation

We implemented Δ -SigningKLPT⁺ and Delfar in SageMath [35]. Our implementation of Δ -SigningKLPT⁺ uses the optimizations described in Appendix D except for Opt. 4. Our implementation is available at

<https://github.com/hiroshi-onuki/DeltaKLPTplus>.

Table 2 shows the running times of KeyGen, Sign, and Verify of Delfar for a ring of n parties. Here, $p_{f,c}$ denotes the prime $c \cdot 2^f - 1$. The running times were measured for $n = 5$ parties as the average over 100 executions with SageMath 10.9 on a Mac mini with an Apple M4 Pro chip and 48 GB of memory. The running times of the key generation and verification procedures are divided by n to report the average running time per party. The running time of the signing procedure is reported as the sum of the running time of the real signing and $n - 1$ times the average running time of the simulator $\mathcal{S}$, where the former is measured

as the running time of the signing procedure with $n = 1$ and the latter is obtained by subtracting the former from the running time of the signing procedure with $n = 5$ and dividing the result by 4.

Table 2. Average running times (in seconds) over 100 executions of Delfar for a ring of n parties, measured for $n = 5$.

p	λ	KeyGen	Sign	Verify
$p_{324.3}$	128	$n \cdot 0.6720$	$5.0528 + (n - 1) \cdot 0.4785$	$n \cdot 0.4502$
$p_{500.27}$	192	$n \cdot 2.2163$	$18.2919 + (n - 1) \cdot 0.8565$	$n \cdot 1.0425$
$p_{664.17}$	256	$n \cdot 4.1499$	$46.3943 + (n - 1) \cdot 2.0513$	$n \cdot 2.1518$

5 Future Work

As a direction for future work, one may ask how small the common norm of equivalent representatives can be. For given ideals I_1 and I_2 , consider pairs $J_1 \sim I_1$ and $J_2 \sim I_2$ satisfying $\text{nrd}(J_1) = \text{nrd}(J_2) = N$. Heuristically, the probability that an ideal of norm $N < p$ exists in a fixed ideal class is of order N/p . If the occurrences in the two ideal classes are modeled as independent, the expected number of common norms up to X is therefore of order $\sum_{N \leq X} (N/p)^2 \asymp X^3/p^2$. The threshold at which this expectation becomes constant is $X \asymp p^{2/3}$. The present `IdealNormReduce` procedure guarantees only a common norm below p , leaving a gap to this heuristic threshold. An algorithm closing this gap and achieving common norm $\tilde{O}(p^{2/3})$ would improve the norm bound from $\tilde{O}(p^4)$ to $\tilde{O}(p^3)$; we leave its construction for future work.

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A Definitions of Σ -Protocols and Ring Signatures

A.1 Σ -Protocols

A Σ -protocol is a three-round interactive proof system between a prover $\mathcal{P}$ and a verifier $\mathcal{V}$ for a relation $\mathcal{R}$ [33, 15].

Let $\mathcal{R} \subseteq \mathcal{X} \times \mathcal{W}$ be a hard relation, which means that given $x \in \mathcal{X}$, it is computationally hard to find $w \in \mathcal{W}$ such that $(x, w) \in \mathcal{R}$. For $(x, w) \in \mathcal{R}$, x is called a *statement* and w is a corresponding *witness*. A Σ -protocol for $\mathcal{R}$ consists of four algorithms $(\text{Gen}, P_1, P_2, V)$ and a challenge space $\mathcal{C}$, and proceeds as follows.

- **Key generation.** The prover $\mathcal{P}$ generates a valid pair $(x, w) \leftarrow \text{Gen}(1^\lambda)$ and publishes x .
- **Commitment.** The prover $\mathcal{P}$ on input (x, w) runs $P_1(x, w)$ to generate a commitment com and internal state st , and sends com to the verifier.
- **Challenge.** The verifier samples a random challenge $\text{chl} \xleftarrow{\$} \mathcal{C}$ and sends chl to the prover.
- **Response.** The prover computes $\text{rsp} \leftarrow P_2(x, w, \text{st}, \text{chl})$ and sends rsp to the verifier.
- **Verification.** The verifier accepts if $V(x, \text{com}, \text{chl}, \text{rsp}) = 1$.

In addition, a Σ -protocol must satisfy the following properties.

Correctness. A Σ -protocol is said to be *correct* if for all $(x, w) \in \mathcal{R}$, an honest execution between $\mathcal{P}(x, w)$ and $\mathcal{V}(x)$ is accepted with probability 1.

Special Soundness. A Σ -protocol is said to be *special sound* if there exists a polynomial-time extractor $\mathcal{E}$ such that given two accepting transcripts

$$(\text{com}, \text{chl}, \text{rsp}), \quad (\text{com}, \text{chl}', \text{rsp}')$$

with $\text{chl} \neq \text{chl}'$, the extractor outputs a witness

$$w \leftarrow \mathcal{E}(x, \text{com}, \text{chl}, \text{rsp}, \text{chl}', \text{rsp}')$$

such that $(x, w) \in \mathcal{R}$.

Honest-Verifier Zero-Knowledge (HVZK). A Σ -protocol is said to be *honest-verifier zero-knowledge (HVZK)* if there exists a probabilistic polynomial-time simulator $\mathcal{S}$ that on input x outputs transcripts $(\text{com}, \text{chl}, \text{rsp})$ whose distribution is computationally indistinguishable from the transcripts generated by an honest interaction between $\mathcal{P}(x, w)$ and an honest verifier $\mathcal{V}(x)$.

A Σ -protocol is also said to be *special HVZK* if there exists a simulator $\mathcal{S}$ that additionally takes chl as input and outputs (com, rsp) such that $(\text{com}, \text{chl}, \text{rsp})$ is indistinguishable from a real transcript. Furthermore, we call this property *special strong HVZK* if the indistinguishability holds even when the distinguisher is given the randomness used by Gen to generate (x, w) .

A.2 Ring Signatures

A ring signature scheme allows a signer to anonymously sign a message on behalf of a set of public keys called a *ring*. The verifier is convinced that the signature was produced by one of the ring members but cannot identify which one [31]. Let n denote the size of the ring. A ring signature scheme consists of the following algorithms.

- $\text{KeyGen}(1^\lambda) \rightarrow (\text{pk}, \text{sk})$: Generates a public/secret key pair.
- $\text{Sign}(\text{sk}_i, m, R) \rightarrow \sigma$: Given a secret key sk_i corresponding to a public key pk_i , a message m , and a ring $R = (\text{pk}_1, \dots, \text{pk}_n)$ containing pk_i , outputs a ring signature σ .
- $\text{Verify}(m, R, \sigma) \rightarrow \{0, 1\}$: Outputs 1 if σ is a valid signature on message m with respect to ring R .

In addition, a ring signature scheme must satisfy the following properties; our security notions follow the standard treatment of ring signatures with adversarially generated and corrupted keys [9].

Correctness. A ring signature scheme is said to be *correct* if for all $(\text{pk}_i, \text{sk}_i)$ generated by KeyGen , all messages m , and all rings R containing pk_i , it holds

$$\Pr[\text{Verify}(m, R, \sigma) = 1] = 1,$$

where $\sigma \leftarrow \text{Sign}(\text{sk}_i, m, R)$.

Unforgeability. A ring signature scheme is said to be *unforgeable with respect to insider corruption* if any probabilistic polynomial-time (PPT) adversary $\mathcal{A}$ has negligible probability of winning the following game played against a challenger.

1. The challenger generates key pairs $(\text{pk}_i, \text{sk}_i) \leftarrow \text{KeyGen}(1^\lambda; r_i)$ for $i \in [n]$ using randomness r_i and sends $\text{PK} = \{\text{pk}_i\}_{i \in [n]}$ to $\mathcal{A}$.
2. The challenger initializes two empty sets S and C .
3. The adversary $\mathcal{A}$ is given access to the following two oracles and may query them an arbitrary polynomial number of times:
 - $\text{OSign}(i, m, R)$: If $i \in [n]$ and $\text{pk}_i \in R$, it returns $\sigma \leftarrow \text{Sign}(\text{sk}_i, m, R)$ and adds (i, m, R) to S .
 - $\text{OCorrupt}(i)$: If $i \in [n]$, it returns r_i and adds pk_i to C .
4. The adversary $\mathcal{A}$ outputs (m^*, R^*, σ^*) . If $R^* \subseteq \text{PK} \setminus C$, $(-, m^*, R^*) \notin S$, and $\text{Verify}(m^*, R^*, \sigma^*) = 1$, then $\mathcal{A}$ wins the game.

Anonymity. A ring signature scheme is said to be *anonymous* if any PPT adversary $\mathcal{A}$ has negligible advantage in the following game played against a challenger.

1. The challenger generates key pairs $(\text{pk}_i, \text{sk}_i) \leftarrow \text{KeyGen}(1^\lambda; r_i)$ for $i \in [n]$ using randomness r_i and sends $\text{PK} = \{\text{pk}_i\}_{i \in [n]}$ to $\mathcal{A}$.
2. The adversary $\mathcal{A}$ is given access to the following signing oracle and may query it an arbitrary polynomial number of times:
 - $\text{OSign}(i, m, R)$: If $i \in [n]$ and $\text{pk}_i \in R$, it returns $\sigma \leftarrow \text{Sign}(\text{sk}_i, m, R)$
3. The adversary $\mathcal{A}$ outputs (i_0, i_1, m, R) such that $i_0, i_1 \in [n]$, $i_0 \neq i_1$, and $\text{pk}_{i_0}, \text{pk}_{i_1} \in R$.
4. The challenger samples a random bit $b \xleftarrow{\$} \{0, 1\}$.
5. The challenger computes $\sigma^* \leftarrow \text{Sign}(\text{sk}_{i_b}, m, R)$ and sends σ^* to $\mathcal{A}$.
6. The adversary $\mathcal{A}$ outputs b^* . The advantage of $\mathcal{A}$ is defined as $\text{Adv}(\mathcal{A}) := |\Pr[b^* = b] - 1/2|$.

In addition, a ring signature scheme is said to provide *full anonymity* if every PPT adversary $\mathcal{A}$ still has negligible advantage in this game when the challenger also sends $\{r_i\}_{i \in [n]}$ to $\mathcal{A}$ in Step 1.

B Explanations for Heuristics 2, 12, and 17

In this section, we give explanations for Heuristics 2, 12, and 17.

B.1 Explanations for Heuristics 2 and 17

We first explain the two heuristics concerning `FullStrongApproximation`, namely Heuristics 2 and 17. For convenience, we restate both heuristics below before giving their explanations. In addition, we give a concrete algorithm for `FullStrongApproximation` in Algorithm 4.

Heuristic 2. Let N be a prime such that $N \neq p, \ell$ and $\omega > 0$. Suppose that ℓ^e, N, C, D satisfy $\left(\frac{p\ell^e(C^2+D^2)}{N}\right) = 1$. If

$$\ell^e > \frac{3 \log(2)}{2\pi} \omega \cdot pN^3 \log(pN^3),$$

then $\text{FullStrongApproximation}_{\ell^e}(N, C, D)$ succeeds with probability at least $1 - 2^{-\omega}$ and runs in expected time polynomial in $\log p$, $\log N$, and ω .

Heuristic 17. Let N and ℓ be primes such that $N \neq p, \ell$, and suppose that ℓ^e, N, C, D satisfy $\left(\frac{p\ell^e(C^2+D^2)}{N}\right) = 1$. Let $\beta_1 \in \mathcal{O}_0$ such that $\beta_1 \notin \ell\mathcal{O}_0$. If

$$\ell^e > \frac{3 \log(2)}{2\pi} \frac{\ell^2 + \ell - 1}{\ell^2 - 1} \omega \cdot pN^3 \log(pN^3),$$

then the modified algorithm $\text{FullStrongApproximation}_{\ell^e}(N, C, D, \beta_1)$ returns an element α such that $\alpha\beta_1 \notin \ell\mathcal{O}_0$ with probability at least $1 - 2^{-\omega}$, and runs in expected time polynomial in $\log p$, $\log N$, and ω .

Explanation for Heuristic 2. Set $R := \frac{4\ell^e}{pN^2}$. The algorithm searches for x, y and $v \in \mathbb{Z}^2$ satisfying

$$x^2 + y^2 = pR - p\|Av - v_T\|^2.$$

For every $v \in \mathbb{Z}^2$, the right-hand side is an integer. By the Gaussian heuristic, the number of vectors satisfying $\|Av - v_T\|^2 < R$ is

$$K \approx \frac{\pi \cdot R}{\det(A)} = \frac{4\pi\ell^e}{pN^3}.$$

We assume that, when v is sampled uniformly from the K candidate vectors, m_v behaves as a uniformly random positive integer at most pR . By the prime number theorem for arithmetic progressions, m_v is prime and congruent to 1 modulo 4 with probability approximately $1/(2 \log pR)$. In this case, we can find x, y satisfying $x^2 + y^2 = m_v$ via Cornacchia's algorithm. We further assume that the parity constraint $(x, y) \equiv (t, z) \pmod{2}$ holds independently with probability $2/3$ as discussed in [24, Section 3.1.3]. Thus, one candidate succeeds with probability approximately $1/(3 \log pR)$. Consequently,

$$\begin{aligned} \Pr[\text{FullStrongApproximation}_{\ell^e}(N, C, D) \text{ succeeds}] &\approx 1 - \left(1 - \frac{1}{3 \log pR}\right)^K \\ &\geq 1 - \exp\left(-\frac{K}{3 \log pR}\right) \\ &\approx 1 - \exp\left(-\frac{4\pi\ell^e}{3pN^3 \log pR}\right) \\ &\geq 1 - \exp\left(-\frac{4\pi\ell^e}{3pN^3 \log(\ell^e)}\right). \end{aligned}$$

Algorithm 4: FullStrongApproximation

Input: a prime power ℓ^e , a prime N , integers C, D such that $\left(\frac{p\ell^e(C^2+D^2)}{N}\right) = 1$, and optionally $\beta_1 \in \mathcal{O}_0$ (default: $\beta_1 = \perp$).

Output: $\alpha \in \mathcal{O}_0$ such that $\alpha \equiv u(C\mathbf{j} + D\mathbf{k}) \pmod{N}$ for some $u \in \mathbb{Z}$ and $\text{nrd}(\alpha) = \ell^e$.

- 1 $u \leftarrow$ a square root of $\frac{4\ell^e}{p(C^2+D^2)}$ modulo N , choosing an even representative;
- 2 $v_0 \leftarrow (z_0, t_0) \in \mathbb{Z}^2$ such that $2pu(Cz_0 + Dt_0) \equiv \frac{4\ell^e - p(C^2+D^2)u^2}{N} \pmod{N}$;
- 3 $v_T \leftarrow -v_0 - \frac{u}{N}(C, D)$;
- 4 $a \leftarrow -C/D \pmod{N}$;
- 5 $A \leftarrow \begin{pmatrix} 1 & 0 \\ a & N \end{pmatrix}$;
- 6 $S \leftarrow \left\{ v \in \mathbb{Z}^2 \setminus \{0\} \mid \|Av - v_T\|^2 < \frac{4\ell^e}{pN^2} \right\}$;
- 7 **for** $v \in S$ **do**
- 8 $m_v \leftarrow \frac{4\ell^e}{N^2} - p\|Av - v_T\|^2$;
- 9 **if** m_v is a prime congruent to 1 modulo 4 **then**
- 10 // Cornacchia's algorithm
- 10 $(x, y) \leftarrow$ integers such that $x^2 + y^2 = m_v$;
- 11 **else**
- 12 **continue**;
- 13 $(z, t) \leftarrow Av + v_0$;
- 14 **if** $(x, y) \equiv (t, z) \pmod{2}$ **then**
- 15 $\mu \leftarrow \frac{x-t}{2} + \frac{y-z}{2}\mathbf{i} + z\frac{1+\mathbf{j}}{2} + t\frac{1+\mathbf{k}}{2}$;
- 16 $\alpha \leftarrow \frac{u}{2}(C\mathbf{j} + D\mathbf{k}) + N\mu$;
- 17 **if** $\beta_1 = \perp$ **or** $\alpha\beta_1 \notin \ell\mathcal{O}_0$ **then**
- 18 **return** α ;

The last expression is at least $1 - 2^{-\omega}$ whenever

$$\frac{\ell^e}{\log(\ell^e)} > \frac{3 \log(2)}{4\pi} \omega p N^3.$$

This holds when

$$\ell^e > \frac{3 \log(2)}{2\pi} \omega p N^3 \log(p N^3).$$

Explanation for Heuristic 17. We now explain Heuristic 17. Fix an $\mathbb{F}_\ell$ -algebra isomorphism

$$\rho_\ell : \mathcal{O}_0 / \ell\mathcal{O}_0 \xrightarrow{\sim} M_2(\mathbb{F}_\ell),$$

and write $A = \rho_\ell(\alpha + \ell\mathcal{O}_0)$ and $B = \rho_\ell(\beta_1 + \ell\mathcal{O}_0)$. Since the reduced norm corresponds to the determinant under ρ_ℓ and $\text{nrd}(\alpha) = \ell^e$, we have $\det(A) = 0$, so A is singular. We assume that A is uniformly distributed over the singular matrices. Since $\beta_1 \notin \ell\mathcal{O}_0$, the matrix B is nonzero. The additional condition $\alpha\beta_1 \notin \ell\mathcal{O}_0$ is equivalent to $AB \neq 0$. There are $\ell(\ell^2 + \ell - 1)$ singular matrices in $M_2(\mathbb{F}_\ell)$.

We distinguish two cases. If B is invertible, then $AB = 0$ if and only if $A = 0$. Hence, the condition $AB \neq 0$ holds with probability

$$1 - \frac{1}{\ell(\ell^2 + \ell - 1)}.$$

If B is singular, then, since $B \neq 0$, it has rank one. In this case, there are exactly ℓ^2 such matrices A satisfy $AB = 0$. Therefore, the condition $AB \neq 0$ holds with probability

$$\frac{\ell(\ell^2 + \ell - 1) - \ell^2}{\ell(\ell^2 + \ell - 1)} = \frac{\ell^2 - 1}{\ell^2 + \ell - 1}.$$

Consequently, regardless of whether B is singular or invertible, the condition holds with probability at least $\frac{\ell^2 - 1}{\ell^2 + \ell - 1}$.

Thus, the success probability from Heuristic 2 is reduced by this factor, which gives the stated bound by the same argument. The additional test is computable in polynomial time and hence does not change the expected running-time estimate.

B.2 Explanation for Heuristic 12

We next explain Heuristic 12, which we first restate for convenience.

Heuristic 12. *Let N be a positive integer prime to p , and sample left $\mathcal{O}_0$ -ideals J_1 and J_2 uniformly at random from the set of left $\mathcal{O}_0$ -ideals of norm N such that there is no prime q for which both J_1/q and J_2/q are contained in $\mathcal{O}_0$. Then, the probability that p divides $\text{nrd}(x)$ in $\text{IdealNormReduce}(J_1, J_2)$ is in $O(1/p)$. Conditioned on $p \nmid \text{nrd}(x)$, the probability that p divides $\text{nrd}(\beta_1)$ is in $O(1/p^2)$.*

Explanation for Heuristic 12. If $N < p$, then $\text{nrd}(x)$ and $\text{nrd}(\beta_1)$ are prime to p , since $\text{nrd}(x) \leq \delta\sqrt{pN} < p$ and $\text{nrd}(\beta_1)/N \leq \delta^{3/2}p^{3/4}N^{1/4} < p$. We may therefore assume that $N > p$.

Heuristically, the number of endomorphisms of E_0 with norm at most d is approximately $\max\{d, d^2/p\}$. The probability that such an endomorphism y satisfies $J_2 y \subset J_1$ is approximately $1/N$. Consequently, the probability that there exists $y \in \mathcal{O}_0 \cap J_2^{-1}J_1$ with $\text{nrd}(y) \leq d$ is heuristically in $O(\max\{d, d^2/p\}/N)$.

Assume that $p \mid \text{nrd}(x)$. Then there exists $x' \in \mathcal{O}_0$ such that $x = x'\mathbf{j}$, and hence $J_2 x' \subset \mathbf{j}J_1\mathbf{j}^{-1}$. The ideal $\mathbf{j}J_1\mathbf{j}^{-1}$ is the Frobenius conjugate of J_1 and is uniformly random among the left $\mathcal{O}_0$ -ideals of norm N . The probability that $\mathcal{O}_0 \cap J_2^{-1}\mathbf{j}J_1\mathbf{j}^{-1}$ contains a nonzero element of norm at most $\delta\sqrt{N/p}$ is in $O(\max\{1/\sqrt{pN}, 1/p^2\})$. Since $N > p$, this gives the first part of Heuristic 12.

For the second part, assume that $p \nmid \text{nrd}(x)$. Since $J_1 \cap x^{-1}J_2x = J_1 \cap (x/n)^{-1}J_2(x/n)$ for every nonzero integer n , we may assume that x is primitive. If $p \mid \text{nrd}(\beta_1)$, then $\beta'_1 := \mathbf{j}^{-1}\beta_1 \in J_1$, because N is prime to p . Let $x' := \mathbf{j}^{-1}x\mathbf{j}$. We then have $\beta'_1 \in x'^{-1}J_2x$.

Set $J'_1 := \chi_{J_1}(\beta'_1)$ and $J := \beta'_1 J'_1{}^{-1}J_2x\beta'_1{}^{-1}$. Since $J_2x \subset J_1$ and $\mathcal{O}_R(J'_1) = \beta'_1\mathcal{O}_R(J_1)\beta'_1{}^{-1}$, the lattice J is a left $\mathcal{O}_R(J'_1)$ -ideal of norm $\text{nrd}(x)$. The condition

$\beta'_1 \in x'^{-1}J_2x$ is equivalent to $\beta'_1x^{-1}J_2^{-1} \subset x'^{-1}\mathcal{O}_0$. By taking the conjugates of both sides, we obtain $\text{nrd}(J'_1)J_2x\beta_1'^{-1} \subset \mathcal{O}_0x'$, where the left-hand side is equal to J'_1J . Heuristically, the probability that a connecting ideal from $\mathcal{O}_0$ to $\mathcal{O}_R(J'_1)$ of a fixed norm satisfies $J'J \subset \mathcal{O}_0x'$ is approximately $1/\text{nrd}(x)$ because, roughly speaking, this means the push-forward of $\mathcal{O}_0x'$ by J' is J . Since the number of left $\mathcal{O}_0$ -ideals of norm at most d is in $O(d^2)$ and approximately $1/p$ of them are right $\mathcal{O}_R(J'_1)$ -ideals, the probability that a random such ideal J' of norm at most d satisfies this inclusion is in $O(d^2/(p\text{nrd}(x)))$. Finally, $\text{nrd}(J'_1) = \text{nrd}(\beta_1)/(pN) \leq \delta\sqrt{\text{nrd}(x)/p}$, which heuristically gives probability $O(1/p^2)$.

Remark 22. Another way to view the heuristic is to regard $\mathcal{O}_0 \cap J_2^{-1}\mathbf{j}J_1\mathbf{j}^{-1}$ and $J_1 \cap x'^{-1}J_2x$ as Haar-random lattices scaled according to their covolumes. Divisibility of $\text{nrd}(x)$ or $\text{nrd}(\beta_1)$ by p implies the existence of a nonzero element whose norm is approximately a factor $1/p$ smaller than the Minkowski bound. For a Haar-random lattice, the probability that such an element exists is in $O(1/p^2)$; see, for example, [5, Theorem 1]. The difference between the probabilities $O(1/p)$ and $O(1/p^2)$ above is due to the special structure of $\mathcal{O}_0 \cap J_2^{-1}\mathbf{j}J_1\mathbf{j}^{-1}$ inherited from $\mathcal{O}_0$. This difference disappears when $N > p^3$.

Remark 23. In Algorithm 2, the ideals $J_1 = L_1\gamma$ and $J_2 = L_2\bar{\gamma}$ obtained immediately after the randomization phase are not distributed as uniformly random ideals. In particular, when the common norm of L_1 and L_2 is less than p , the norm of a shortest $x \in \mathcal{O}_0 \cap J_2^{-1}J_1$ is divisible by p with high probability. This phenomenon does not occur in our setting, where the common norm of L_1 and L_2 is approximately p^3 . See Appendix D.2 for details.

C Omitted Proofs

For a prime q and a lattice L in $\mathcal{O}_0$, write $L_q := L \otimes_{\mathbb{Z}} \mathbb{Z}_q$.

C.1 Proof of Lemma 5

Lemma 5. *Let I_1 and I_2 be left $\mathcal{O}_0$ -ideals of the same norm N . Then, $\mathcal{O}_0 \cap I_2^{-1}I_1$ is a lattice in $\mathcal{O}_0$ and $[\mathcal{O}_0 : \mathcal{O}_0 \cap I_2^{-1}I_1] \leq N$.*

Proof. It is clear that $\mathcal{O}_0 \cap I_2^{-1}I_1$ is a lattice in $\mathcal{O}_0$.

From the local-global principle [37, Lemma 9.6.7], it suffices to show that

$$[(\mathcal{O}_0)_q : (\mathcal{O}_0 \cap I_2^{-1}I_1)_q] \leq q^{v_q(N)},$$

where $v_q(N)$ is the q -adic valuation of N . If q does not divide N , then the inequality is clear since $(I_1)_q = (I_2)_q = (\mathcal{O}_0)_q$. If $q = p$, then $(I_1)_p = (I_2)_p = \mathfrak{P}^{v_p(N)}$, where $\mathfrak{P}$ is the unique maximal ideal of $(\mathcal{O}_0)_p$. Therefore, we have $[(\mathcal{O}_0)_p : (\mathcal{O}_0 \cap I_2^{-1}I_1)_p] = 1$, and the inequality holds.

Let $q \neq p$ be a prime dividing N , and $v := v_q(N)$. Then, we have $(\mathcal{O}_0)_q \cong M_2(\mathbb{Z}_q)$. Under this isomorphism, we can write

$$(I_1)_q = M_2(\mathbb{Z}_q)m_1, \quad (I_2)_q = M_2(\mathbb{Z}_q)m_2,$$

where $m_1, m_2 \in M_2(\mathbb{Z}_q)$ are matrices of determinant q^v . By the theory of elementary divisors, there exist $u_1, u_2, w_1, w_2 \in GL_2(\mathbb{Z}_q)$ and integers $0 \leq f_1, f_2 \leq v$ such that, for $i \in \{1, 2\}$,

$$d_i = \begin{pmatrix} q^{f_i} & 0 \\ 0 & q^{v-f_i} \end{pmatrix}$$

and

$$m_1 = u_1 d_1 w_1, \quad m_2 = u_2 d_2 w_2.$$

Therefore, we have

$$\begin{aligned} (\mathcal{O}_0 \cap I_2^{-1} I_1)_q &\cong \{\mu \in M_2(\mathbb{Z}_q) \mid m_2 \mu m_1^{-1} \in M_2(\mathbb{Z}_q)\} \\ &= \{\mu \in M_2(\mathbb{Z}_q) \mid u_2 d_2 w_2 \mu w_1^{-1} d_1^{-1} u_1^{-1} \in M_2(\mathbb{Z}_q)\}. \end{aligned}$$

This implies that $\mu \in (\mathcal{O}_0 \cap I_2^{-1} I_1)_q$ if and only if

$$w_2 \mu w_1^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

where $a, b, c, d \in \mathbb{Z}_q$ satisfy $v_q(a) \geq f_1 - f_2$, $v_q(b) \geq v - f_1 - f_2$, $v_q(c) \geq f_1 + f_2 - v$, and $v_q(d) \geq f_2 - f_1$. Therefore, we have

$$[(\mathcal{O}_0)_q : (\mathcal{O}_0 \cap I_2^{-1} I_1)_q] = q^{|f_1 - f_2| + |v - f_1 - f_2|}.$$

If $f_1 + f_2 \leq v$, then $|f_1 - f_2| + |v - f_1 - f_2| = v - 2 \min(f_1, f_2) \leq v$. If $f_1 + f_2 > v$, then $|f_1 - f_2| + |v - f_1 - f_2| = 2 \max(f_1, f_2) - v \leq v$. In either case, we have $|f_1 - f_2| + |v - f_1 - f_2| \leq v$. $\square$

C.2 Proof of Lemma 6

Lemma 6. *Let I_1 and I_2 be left $\mathcal{O}_0$ -ideals of the same norm N , and $x \in \mathcal{O}_0 \cap I_2^{-1} I_1$ be a nonzero element. Then, we have $[\mathcal{O}_0 : I_1 \cap x^{-1} I_2 x] \leq N^2 \cdot \text{nrd}(x)$.*

Proof. Since $[\mathcal{O}_0 : I_1] = N^2$, the claim is equivalent to $[I_1 : I_1 \cap x^{-1} I_2 x] \leq \text{nrd}(x)$. By the local-global principle, it suffices to show that

$$[(I_1)_q : (I_1 \cap x^{-1} I_2 x)_q] \leq q^{v_q(\text{nrd}(x))}.$$

In the case where $q = p$, we have $(I_1)_p$ and $(x^{-1} I_2 x)_p$ are equal to $\mathfrak{P}^{v_p(N)}$ as in the proof of Lemma 5. Therefore, we have $[(I_1)_p : (I_1 \cap x^{-1} I_2 x)_p] = 1$.

Let q be a prime not dividing $\text{nrd}(x)$. Then, x is a unit in $(\mathcal{O}_0)_q$, and we have $(x^{-1} I_2 x)_q = (I_2)_q$ since $I_2 x \subset I_1$. Therefore, $[(I_1)_q : (I_1 \cap x^{-1} I_2 x)_q] = 1$.

Let $q \neq p$ be a prime dividing $\text{nrd}(x)$, and $v := v_q(\text{nrd}(x))$. As in the proof of Lemma 5, we identify $(\mathcal{O}_0)_q$ with $M_2(\mathbb{Z}_q)$ and write

$$(I_1)_q = M_2(\mathbb{Z}_q) m_1, \quad (I_2)_q = M_2(\mathbb{Z}_q) m_2.$$

We set $z := m_2 x m_1^{-1}$. Since $I_2 x \subset I_1$, we have $z \in M_2(\mathbb{Z}_q)$. In addition, we have $v_q(\det z) = v$.

By the theory of elementary divisors, there exist $u_1, u_2, w_1, w_2 \in \text{GL}_2(\mathbb{Z}_q)$ and integers $0 \leq a, b \leq v$ such that

$$x = u_1 \begin{pmatrix} q^a & 0 \\ 0 & q^{v-a} \end{pmatrix} w_1, \quad z = u_2 \begin{pmatrix} q^b & 0 \\ 0 & q^{v-b} \end{pmatrix} w_2.$$

Then, we have

$$\begin{aligned} (I_1 \cap x^{-1} I_2 x)_q &\cong \{\mu m_1 \mid \mu \in M_2(\mathbb{Z}_q), x \mu m_1 x^{-1} \in M_2(\mathbb{Z}_q) m_2\} \\ &= \{\mu m_1 \mid x \mu z^{-1} \in M_2(\mathbb{Z}_q)\} \\ &= \left\{ \mu m_1 \mid \begin{pmatrix} q^a & 0 \\ 0 & q^{v-a} \end{pmatrix} w_1 \mu w_2^{-1} \begin{pmatrix} q^{-b} & 0 \\ 0 & q^{b-v} \end{pmatrix} \in M_2(\mathbb{Z}_q) \right\}. \end{aligned}$$

By the same argument as in the proof of Lemma 5, we have $[(I_1)_q : (I_1 \cap x^{-1} I_2 x)_q] \leq q^v$. $\square$

D Optimizations for $\Delta\text{-KLPT}^+$

In this section, we discuss several optimizations for $\Delta\text{-KLPT}^+$.

The dominant cost of $\Delta\text{-KLPT}^+$ is the repeated calls to `IdealNormReduce`. Although the expected number of calls is in $O(\log p \log \log p)$ under Heuristic 14, this number can be large in practice. More precisely, combining this heuristic with the bound in Proposition 8 gives the following upper bound on the expected number of `IdealNormReduce` calls:

$$4 \log p \left\lceil \frac{1}{2} \log_2 \left(\log_2 \left(\frac{2^{e_0} \text{nrd}(\gamma)}{p} \right) + C_1 \right) + C_2 \right\rceil, \quad (2)$$

where C_1 and C_2 are constants in Proposition 8, e_0 is the exponent in Heuristic 3, and γ is the randomization quaternion in Algorithm 2.

For example, for $p = p_{324.3}$, the bound in Proposition 8 is 7, and the expected total number of `IdealNormReduce` calls is approximately 6319 by approximating e_0 by $3 \log_2 p$ and $\text{nrd}(\gamma)$ by p^2 .

This section gives the following four practical optimizations that can reduce the expected number of `IdealNormReduce` calls.

- Opt. 1 Reducing the common norm before the randomization phase.
- Opt. 2 Reducing the norm of the randomization quaternion γ .
- Opt. 3 Enumerating candidates in the final `IdealNormReduce` call.
- Opt. 4 Mitigating the condition on the final common norm N .

For our implementation of the ring signature scheme Delfar, we use these optimizations except for Opt. 4. The rationale for and details of these optimizations are explained in the following subsections.

D.1 Reducing the Number of IdealNormReduce Calls

1. Reducing the Norm before the Randomization Phase. Opt. 1 is to reduce the common norm of the input ideals before multiplying by a randomization quaternion γ .

The reduction before the randomization phase requires at most the bound in Proposition 8 calls to IdealNormReduce. Since this reduction makes the common norm less than p , the upper bound on the expected number of IdealNormReduce calls after the randomization phase is obtained by replacing 2^{e_0} in Eq. (2) with p . As explained in Appendix D.2, this optimization should be used with Opt. 2 to avoid the situation where the lattice $\mathcal{O}_0 \cap I_2^{-1}I_1$ has a small element whose norm is divisible by p .

Remark 24. Alternatively, we can reuse an ideal pair failed at the check in line 15 of Algorithm 2 as the input of the next iteration. This change slightly decreases the expected number of IdealNormReduce calls compared with Opt. 1. However, the distribution of the invariant $\Psi_{L_1, L_2}(J_1, J_2)$ of the ideal pair (J_1, J_2) may not be uniform in the set of possible values of $\Psi_{L_1, L_2}(J_1, J_2)$ since the starting ideal pair depends on the order of randomization quaternions γ . Therefore, we do not use this optimization in our implementation.

2. Reducing the Norm of the Randomization Quaternion. The invariant $\Psi_{L_1, L_2}(J_1, J_2) = \theta(\gamma^2)$ in Algorithm 2 is preserved under replacement of γ by $a\gamma + b\mathbf{p}\mathbf{i}$ for any integers a and b with $p \nmid a$. This allows us to replace γ by a shortest vector in the lattice spanned by γ and $\mathbf{p}\mathbf{i}$. This change reduces the norm of γ to at most approximately p , and hence reduces the expected number of IdealNormReduce calls in Eq. (2).

Remark 25. We can further reduce the norm of γ by sampling γ from the circle of radius R in $\mathbb{Z}[\mathbf{i}]$ for some $R < p$. For example, if we can take $R = 4 \cdot 2^\lambda \log p$ then we expect the number of possible outputs of $\Delta\text{-KLPT}^+$ to be approximately 2^λ , and hence the full anonymity of Delfar is preserved. In addition, we can take $R \in O(\log p)$ if we only need to find a single output of $\Delta\text{-KLPT}^+$. However, we do not use this optimization in our implementation since the resulting reduction in the expected number of IdealNormReduce calls is modest.

3. Enumerating Candidates in the Final IdealNormReduce Call. In the final IdealNormReduce call in line 13 of Algorithm 2, we can enumerate elements β_1 in the lattice L_x satisfying $\text{nrd}(\beta_1)/\text{nrd}(I_1) < p$. Selecting β_1 from this enumerated set such that $\text{nrd}(\beta_1)/\text{nrd}(I_1)$ is prime congruent to 3 modulo 4 if such an element exists increases the probability of breaking the outer randomization loop.

More precisely, if the input of IdealNormReduce satisfies $\delta^{3/2}p^{3/4}\text{nrd}(I_1)^{1/4} < p$ then we change line 5 of Algorithm 3 as follows: We enumerate at most $4 \log p$ elements β_1 in L_x satisfying $\text{nrd}(\beta_1)/\text{nrd}(I_1) < p$ and select β_1 such that $\text{nrd}(\beta_1)/\text{nrd}(I_1)$ is prime congruent to 3 modulo 4 if such an element exists, otherwise we select a shortest nonzero element of L_x .

4. Mitigating the Condition on the Final Common Norm. We can mitigate Condition 2 in (1), the common norm N is prime congruent to 3 modulo 4, by also allowing primes congruent to 1 modulo 4. `IdealModConstraint` may fail for a prime N congruent to 1 modulo 4. However, the probability of failure is negligible for sufficiently large N , in particular at most $\frac{4}{N+1}$ (see Section 4.3 in [23]). Although this probability is negligible in most cases, this optimization changes the success probability of Δ -KLPT⁺. For simplicity, we therefore do not use this optimization in our implementation.

Furthermore, we may also allow N to be a composite number with a few prime factors and known factorization. The reason we require N to have a few prime factors is that many prime factors decrease the probability of success of `IdealModConstraint` and the probability that the condition on the Legendre symbol (Condition 3 in (1)) holds. We require the factorization of N to compute a square root modulo N in `FullStrongApproximation`. The probability that a random composite number N satisfies these conditions is low, so the effect of this optimization may be limited.

D.2 Special Structure of the Ideals $L_1\gamma$ and $L_2\bar{\gamma}$

In this subsection, we note that Opt. 1 should be used with Opt. 2 to avoid the situation where the lattice $\mathcal{O}_0 \cap I_2^{-1}I_1$ has a small element whose norm is divisible by p .

The ideals $L_1\gamma$ and $L_2\bar{\gamma}$ in Algorithm 2 cannot be regarded as uniformly random left $\mathcal{O}_0$ -ideals with norm $2^{e_0} \text{nr}d(\gamma)$. Therefore, we cannot directly apply Heuristic 12 to these ideals. In the following, we show that the lattice $\mathcal{O}_0 \cap I_2^{-1}I_1$ for $I_1 = L_1\gamma$ and $I_2 = L_2\bar{\gamma}$ has a small element whose norm is divisible by p in a certain case.

Proposition 26. *Let L_1 and L_2 be left $\mathcal{O}_0$ -ideals of the same norm N , and let γ be a quaternion in $\mathbb{Z}[\mathbf{i}]$. We denote by L the lattice $\mathcal{O}_0 \cap (L_2\bar{\gamma})^{-1}(L_1\gamma)$. Then, there exists a nonzero element β in the lattice $L \cap \mathbf{j}\mathbb{Z}[\mathbf{i}]$ such that $\text{nr}d(\beta) \leq \frac{4}{\pi}pN$.*

Proof. Let L' be the lattice $\mathcal{O}_0 \cap L_2^{-1}L_1$. From Lemma 5 and $[\mathcal{O}_0 : L'] \geq [\mathbf{j}\mathbb{Z}[\mathbf{i}] : L' \cap \mathbf{j}\mathbb{Z}[\mathbf{i}]]$, we have $[\mathbf{j}\mathbb{Z}[\mathbf{i}] : L' \cap \mathbf{j}\mathbb{Z}[\mathbf{i}]] \leq N$. Therefore, by Minkowski's bound, there exists a nonzero element $\beta' \in L' \cap \mathbf{j}\mathbb{Z}[\mathbf{i}]$ such that $\text{nr}d(\beta') \leq \frac{4}{\pi}pN$.

Let $\beta = \bar{\gamma}^{-1}\beta'\gamma$. Then we have $\beta \in (L_2\bar{\gamma})^{-1}(L_1\gamma)$. Since $\gamma \in \mathbb{Z}[\mathbf{i}]$, we have $\beta'\gamma = \bar{\gamma}\beta'$. This implies that $\beta = \beta'$, and hence $\beta \in \mathcal{O}_0$. Therefore, we have $\beta \in L \cap \mathbf{j}\mathbb{Z}[\mathbf{i}]$. This proves the proposition. $\square$

An element β in the above proposition may be a short element in the lattice L when $\frac{4}{\pi}pN \leq \delta\sqrt{pN \text{nr}d(\gamma)}$, i.e., when $\text{nr}d(\gamma) \gtrsim pN$. In Algorithm 2, this never happens since we have $\text{nr}d(\gamma) < p^2$ and $N \gtrsim p^3$. However, if we use Opt. 1, then we have $N < p$ and this condition is satisfied for γ with $\text{nr}d(\gamma) \approx p^2$. We can avoid this situation by using Opt. 2 since it reduces $\text{nr}d(\gamma)$ to at most p .

D.3 Effect of the Optimizations

By approximating e_0 by $3 \log_2 p$ and $\text{nrd}(\gamma)$ by p^2 in (2) and ignoring the constants C_1 and C_2 , the bound on the expected number of `IdealNormReduce` calls is approximately $2 \log p (\log_2 \log_2 p + 2)$. By applying Opts. 1 and 2, the bound is reduced to approximately

$$2 \log p \log_2 \log_2 p + \frac{1}{2} (\log_2 \log_2 p + 1).$$

Thus, we can reduce the expected number of `IdealNormReduce` calls by approximately $4 \log p$. It is difficult to estimate the effect of Opt. 3 since it depends on the common norm of the input ideals for the final `IdealNormReduce` call.

To see the effect of these optimizations in practice, we implemented $\Delta\text{-KLPT}^+$ with and without the optimizations and counted the number of `IdealNormReduce` calls for different values of p . Our implementation for the experiment is available at our GitHub repository. Table 3 shows the average numbers of calls in 100 executions of $\Delta\text{-KLPT}^+$ for $p \in \{p_{324.3}, p_{500.27}, p_{664.17}\}$.

Table 3. Average numbers of `IdealNormReduce` calls in 100 executions of $\Delta\text{-KLPT}^+$. The first row shows the average numbers of `IdealNormReduce` calls without any optimizations. The second row shows the average numbers of `IdealNormReduce` calls with Opt. 1 and Opt. 2. The third row shows the average numbers of `IdealNormReduce` calls with Opt. 1, Opt. 2, and Opt. 3. The fourth row shows the average numbers of `IdealNormReduce` calls with all four optimizations.

	$p_{324.3}$	$p_{500.27}$	$p_{664.17}$
No optimizations	4,745	10,134	14,254
Opts. 1–2	4,083	7,573	9,926
Opts. 1–3 (used in Delfar)	535	1,377	4,587
Opts. 1–4	261	636	3,016

E Use of AI Tools

We disclose the use of generative AI tools in this work. The tool we used was Claude (Anthropic; Opus 4.8 and 5, and Fable 5 and 5.1). The text of this paper was written by the authors, and the tool was not used to generate any part of it except for editing-level assistance. All AI-assisted material listed below was reviewed, tested, and verified by the authors, who take full responsibility for its correctness.

Implementation. The SageMath implementation at our GitHub repository was developed with substantial AI assistance. In particular, the following files were almost entirely generated by Claude:

- `montgomery.py`: x -only arithmetic on Montgomery curves,
- `lattice.py`: lattice reduction and enumeration,
- `util.py`: utility functions, including a deterministic pseudorandom generator,
- `benchmark.sage` and `count_iter_INR.sage`: the measurement scripts used to produce Tables 2 and 3,
- `README.md`: the documentation of the repository.

The remaining files, which in particular contain the main algorithms proposed in this paper (`IdealNormReduce`, Δ -KLPT⁺, and Δ -SigningKLPT⁺), were written by the authors. For these files, Claude was used for partial assistance, namely debugging, performance optimization, and the implementation of some individual functions. Claude was prompted interactively with natural-language specifications of the functions, error messages, and profiling output; all generated code was reviewed, refactored where necessary, and validated by the authors with the test scripts.

Analysis of the randomization phase. The core idea of the randomization technique in Δ -KLPT⁺, which is explained in Section 3.3, comes from an analysis of the distribution of the outputs of `IdealNormReduce` explored with the help of Claude. In particular, the fact that `IdealNormReduce` preserves the invariant Ψ_{I_1, I_2} (Proposition 11) was discovered by Claude. In addition, the proofs of Lemmas 5 and 6 were obtained with the help of Claude. The formulation of these statements and their proofs as they appear in this paper were written and verified by the authors.